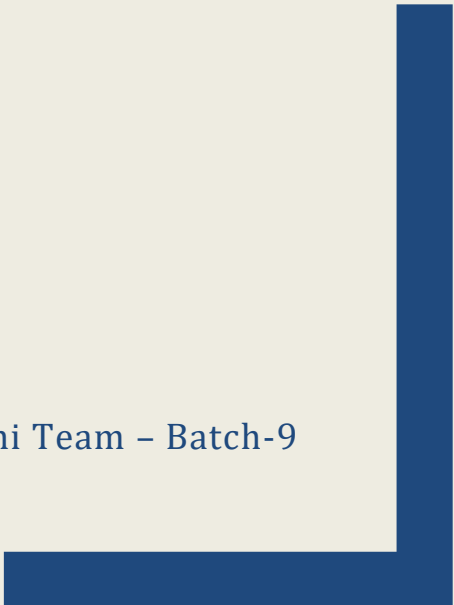




PGDBA – Batch-8

INTERVIEW EXPERIENCE

Alumni Team – Batch-9



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Mastercard D&S

1. Manvinder Chahar

The selection process involved case studies before the interview. I worked on the below case study:

- a. To improve cross-selling & Up-selling for Bank - increase basket size of customers I Made Recommender System
- b. The other one was related to ONDC

Interview Round 1:

I was given a case study whose description is given as below:

Problem Statement: A bank has launched a credit card targeting customers who use credit cards for general purpose transactions. But their existing customer base for this card is not as per their expectation. Suggest Things to client on basis of every minute level of data available. Discussion initiated on:

- a. General Purpose Cards for people who just entered the corporate world. earning 7-12 pa.
- b. Take out this customer segment and analyse on How much they spend, Frequency Analysis on spends via Credit card by this cohort? Since retailer don't accept credit card/ people prefer UPI/

Que. Is your credit card able to satisfy the demands of the customer base?

Que. Identify the needs of the customer & customer base?

Interviewers asked if interviewee has any questions:

1. Que asked by Manvinder: If I appeared for AI Garage. Is that considered a negative point?

Response of interviewer: In turn asked a question to him that is he willing to join?

The interviewee gave priority to AI Garage But is Ready to join MCard

Interview Round 2:

Situation: I was the last one to be interviewed by that panellist: He was Booking a cab in front of the interviewee. Was also asked to give instructions to cab Driver.

Interviewer was highly experienced.

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Following questions were asked:

1. Talk about your Previous Work Experience
 2. What work was done before & during PGDBA?
 3. I have work experience in Tata Motors and was asked, " How can you convince someone who has spent so many years in industry? - Due to mention of Dealer Profitability in WorkEx
Response: Understand Expectation of Dealer Principle: What is the expected number of cash flow: This can help in convincing
 4. What will you do if profitability is not increasing?
 5. Understand the Break-even No.? Explain the following:
 6. What is the trend of Achieving Break-even numbers? If falling, define a new break-even level.
 7. Questions asked about problems faced during Work Experience
 - a. Dealers were Billing the vehicles but not selling the vehicles to get incentives from Tata Motors
 - b. This can be checked by How much Inventory is present? Match it with Sales-Can easily identify.
 - c. How to ask dealers to not do that?
 - d. Since the vehicle is not being delivered, no incentive will be given for that.
 - e. More vehicles will be delivered on basis of bills which will increase the inventory rent cost.
 8. Question on time Series -
 - a. What will be the factors to decide Time Series - Answered ARIMA.
 - b. How to Accommodate Seasonality Factor?
 - c. Explain Scarcity of Economy.
 - d. Seasonality in Automobile is during last quarter.
 - e. What if you apply Sarimax? Predict using that?
-

2. Pranjal Sarin

Round 1:

1. Why DnS when AI garage was your first choice?
2. What is a Decision Tree and Collaborative filtering? (Tip: Don't get into maths as the panellist is not that much interested)

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3. Case: It has been many years since credit card spending has been low and the customer base has not increased. Identify the problem. How do you make it profitable?
4. Why Mastercard DNS?

Company Specific Advice: More of Case Study Prep is required. Explaining concepts in layman terms is the key.

3. Priyanshu [Joined]

Two case studies were given before the interview for shortlisting. Do not remember the details but were related to:

1. ONDC credit card released by Bank:
2. GTM strategy for selective User Cohort

Round 1

Questions Asked:

1. Introduce Yourself:
2. Discussion on Work Experience - Projects done at Axis bank & India Today
3. What Products are Built in India Today & What is the Strategy behind & how is it Scaled?
4. Question related to Axis Bank - What was my basic job - Overview.
5. Case was given with below problem statement:
 - a. A bank is planning to release a Premium Credit card - Do Customer Segmentation. Which Customer to be targeted first?
6. Algorithm was asked - Why K Means ++ in this. What it is?

Tip: Focus more on Business

Interview Lasted for 10-12 minutes.

Round 2:

Felt like Kind of Formality as it was a general discussion over ONDC. Below are the questions asked:

1. Do you know what ONDC does?
2. How will ONDC be better than Amazon?
 - a. Long Chain - Niche Products can be catered.

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- b. Localisation of vendors
- c. One Day Delivery of Amazon Prime will no longer be differentiator.
- 3. Discussion around ONDC better than Amazon & its value chain

General Advice:

- a. Everyone - Develop Soft Skill - Especially in Holidays
 - b. Business Case Competition - Will Help A Lot
 - c. How to Give Introduction - Differentiate Yourself - Have Some Projects Which Are Not Common - Novel idea - Focus on Business Not Just AI / ML
 - d. Reference Book - ISLR
 - e. Write 5 - 6 projects in resume.
-

4. Robin

Round 1:

The interview lasted for around 25 minutes. Following were the questions:

- 1. Tell Me about Yourself | Go Through Your Resume - 50s
Tip: {Have a long and short version of Intro.}
- 2. Why PGDBA after Civil Engineering?
- 3. I was given a Case study:
 - a. There are Multiple Credit Card Segments with client - One of the cards called Platinum Card is not very popular among people. How to increase its reach?
 - b. Response:
Asked Platinum Card Features - Shopping | Who are Target Customers - Upper Middle to Rich | Is there a match between this customer segment and other cards | Clustering & profiling.
- 4. Explain working of decision trees - intuitively
- 5. I was asked to ask a question:
 - a. I asked: What would have been your approach:
 - b. Their Response: Changes in Platinum Card

General Advice:

- a. Coding & ML - Learn it By Heart;
- b. Not much DL but know concepts / definitions.

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- c. ML concepts by heart.
 - d. If technical terms are mentioned in Resume, know them by heart.
 - i. Ex. If Decision trees are mentioned- Why not Lasso, Ridge or any other?
 - e. If Workex is big, most of the things are asked from there. Have a balance of everything.
 - f. Do projects where models can be compared.
-

5. Lagnesh TS:

- 1) The case study was about developing a data-driven platform for personalized next best product recommendations to enhance cross-selling and up-selling opportunities for a Bank.
- 2) I focused on Feature Identification, approaches for Model Building & Evaluation in my presentation.
- 3) No idea.
- 4) The difficulty level was moderate.
- 5) I believe that those who didn't participate in this competition were not shortlisted for the interviews.

Interview Round 1:

- 1. This was a business round.
 - 2. Yes. Your company has built a tool which is used to show the transactions which got approved or rejected for a bank. Many clients have bought that tool but the usage has been on the lower side. How will you try to increase the tool's usage?
 - 3. Why don't you start talking about yourself? Why did you come for PGDBA? Given the above case and asked to solve. What type of role are you looking for? Tell me the difference between research role (AIG) and business-related role (D&S). Which role do you see yourself as fitting? What do you think brings value to business - Research role or solving business problems?
 - 4. The difficulty level was moderate.
-

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6. Ramnath

Interview Round 1:

1. It was a technical round
 2. I was given a case:
 - a. Improve the activation and spending of a credit card. I was asked for an analytics-based approach.
 3. Difficulty of round - This round was Not very hard
-

7. Nikhil Garg

Advice: Case study is average just follow business concepts. Do not go too much technical

8. Parijat Pushpam: [Joined]

Case study:

- 1) Increase the cross selling and upselling of a MasterCard product.
- 2) At the time when companies come you will have to manage classes as well as ppts for multiple companies. I got 6 days of time to research and prepare.
- 3) Presentation should look professional first of all. Every slide should be easy to explain. Check for grammatical mistakes. Use icons. Make clean ppts. If you have won any case competition, then use the template of that PPT. This would save time.
- 4) Not very competitive, but still do your best.
- 5) That is very subjective. Companies filter candidates on multiple criteria which no one would be able to pin point.

Interview Round 1:

- 1) Was this a technical or a business round? - It was a Business round
- 2) Any case study involved - Yes, there was a case study round related to online coaching sites. They wanted me to come up with a forecast for revenue of an offline coaching institute if they want to transform into a fully online unit.
- 3) Some basic ML/SQL questions were also there.
- 4) Difficulty of round - It was fairly competitive.

Interview Round 2:

- 1) Was this a technical or a business round? - Business cum HR round

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- 2) Was case study involved - No case study
- 3) Question asked - Basic HR questions. Not anything new. They wanted to check that whether I would be able to work in the corporate environment, given that I don't have any experience.
- 4) Difficulty of round - It wasn't too difficult. Be calm and give your responses very confidently. You should mean what you say obviously.

Suggestion:

- 1. Be calm. There will be pressure, but still give your best.
 - 2. Be very thorough with your CV.
 - 3. Be very simple and common sensical in your approach.
-

9. Diwakar

I was given a case.:

Case 1: You're an 'amazon go' kinda company, you have a warehouse in your hometown and your market is 100km away in a city named Xyz. Is it best to supply goods to your customer directly from the warehouse and supply your goods from local shops of that city? (Interviewer thought the candidate's experience is from SupplyChain)

10. Nikhil Udayan [Joined]

I had 3 years of work experience in Analytics domain in Health care startup - Excel, Python, PowerBI, SQL)

I was given a **Case study:** (Recommender system for a Bank - No data set is given, just different approaches)

Round 1: (25-30 mins) - 1 panellist

- 1. Past work and projects in past work
- 2. In medical products, what variable did you use and approach? - Past work exp question
- 3. 3 short business cases

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4. Case 1: Digital payment card - how it can beat UPI? Digital payment card vs UPI (Purely business approach was expected)
5. Guesstimates
6. Calculate the weight of an aircraft.
7. How many people will travel from Kolkata to Ranchi daily?

Round 2: 1 panellist

1. Introduction
2. From their resume, they asked them to explain a deep learning project that he did
3. Technical cross questions
4. Asked if he knew SQL
5. Difference between linear regression and logistic regression?
6. Case 1: Master is a bank service provider that provides service to banks. One bank comes to Mastercard and says that their profit is declining heavily. How can Mastercard access this bank? Mastercard has data from other banks, how can you leverage this data? (Solution from him - there can be competitors, we can do competitive analysis, costing analysis if the cost part is affecting the revenue. Using data you can build a classification model to get the characteristics of churning customers)
7. Why Mastercard?

Remark: They're expecting PowerBI and Tableau skills too, case solutions should be small and crisp.

Tips: Work on your communication, prepare case studies, business knowledge is very important, SQL and Python, Go through ISLR and Josh Strammer, Use GPT to learn things

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11. Mukul Mundle

I had 3 years work experience: Assistant Manager - Digital Transformation at Dr. Reddy's Laboratory. Thereafter, worked for 1 year as Business Strategy Manager - Digital Excellence at Jubilant Group)

Pre-interview Case Study:

We (group of 2) had to provide a strategy on upselling & cross-selling of products for a particular bank. We had to describe our strategy and mention what kinds of algorithms we would use. Approach: We had to build a recommender system- content based or collaborative filtering-based system. I explained both these techniques, and I proposed a hybrid solution. We had to submit a ppt. We were given 10-12 days and I spent ~5 hours on it.

Round 1:

It was based on my submitted solution to the case study

We had proposed a hybrid (content+collaborative) solution. Given the profile of customers, we would also do some clustering of customers, do market basket analysis, RFM analysis. Having implemented the solution, we could go for A/B testing. Since a large dataset was involved, I proposed an AWS based solution for implementing this solution.

Their questions: "What's content and collaborative based recommender system, what's A/B testing?"

They asked me about a classification project I had done on insurance claims fraud detection. They enquired about the exact business problem (An insurance company had received various automobile insurance claims and we had to predict whether those claims are fraudulent or not). They asked about various columns in the dataset, how I did data cleaning, feature engineering, how I addressed the imbalance in the data (they asked about SMOTE). They asked about the various models that I had used and finally about the performance metric (which was recall in this case).

Round 2:

Again, questions were asked from the DnS case study.

They asked the same questions. I explained the problem statement, my approach, etc.

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I was further given a case study where mastercard wanted to collaborate with the government and so on which different areas could they collaborate with the government and which kinds of services to the government could they monetize and how (don't recall exactly)

Further Comments: Since I was mostly focused on preparing for DE Shaw, I didn't give enough time to prepare for Mastercard's interview.

12. Amrita Mandal [Joined]

- **Case study:** We were given a case study on customer segmentation and persona building for an online platform and finally went to market strategies for each of the segments.
- I used a 50-50 blend of technical and business. I knew that for any kind of segmentation that has to be done, clustering has to be performed. Building upon that logic made a whole flow from data collection to model building and using the insights to develop business strategies to incentivise each group of customers.
- That was my first company case study, so I took some time to gather my thoughts and prepare the plan. I took around a 7-8 days
- It was moderately competitive. The case study wasn't very tough, with decent knowledge and good ppt skills, you can easily make a compelling presentation.

Round 1 :

1. My round was mostly a conversational round.
2. The interviewer asked me about the same case study and what my thought process was regarding the case study.
3. Rest was more like a culture fit interview.

Remark: It was a good interview. What worked for me I guess was the communication skills and confidence.

13. Lalit Dhakar

- **Case study:** Develop a data-driven platform for personalised recommendations to enhance cross-selling and upselling opportunities for ABC Bank
- One week, It was a segmentation problem had to come up with an end to end solution

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- Presentation Skills, Solution approach
- Similar to the case competitions out there organised by top B -Schools

Round 1 :

1. My round was mostly a conversational round.
2. Tell me something that is not there in your CV
3. Your academic performance has been on a decline from the 10th grade to your B.Tech studies. What do you think might be the reason for this trend?
4. Do you prefer becoming an expert in a specific field or being a jack of all trades?
5. About case study, approach, and reasoning behind that.

Round 2:

1. Walk me through your CV
2. Work ex Projects
3. About work at EXL(Interviewer's last company was EXL before mastercard)
4. De ssstudy: Outline the strategy for introducing a new credit card in Dubai targeted at customers who frequently make everyday purchases.

14. Saurabh Kumar

- Business round, asked around the business of Mastercard
 - Questions asked were very random.
-

15. Utkarsh Sinha

- The case study was about Customer Acquisition Strategy for ONDC based reward credit cards. We had to make a 10 min video explaining our approach to the problem.
- Parameters of evaluation:
- Problem Solving Approach
- Novelty of Solution
- Analytical Technique
- Clarity of Presentation
- The problem was not very difficult. I approached it based on research about similar existing cards.

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Interview Round 1:

1. Case: There is a credit card client who has a platinum segment. The customer base for their platinum segment is very small. They want to increase that customer base. Mastercard has transaction data of their competitors. How do you help your client?
2. Why Mastercard ?
3. You had put AI garage as your first choice. Will you be comfortable working at DNS?

Interview Round 2:

1. Do you think ONDC is a threat or an opportunity for mastercard?
2. Tell me about what mastercard does. Don't give me a wikipedia answer.
3. Any questions you want to ask me?

Tips: Things that may not have worked - needed to do better company research.

15. Abhirup Saha [Joined]:

3 rounds - case study based round

Interview Round 1:

1. It started with intro
2. Main question was: Write down economies of scale shifting the IIMC campus from JOKA to new town (Interview was looking for figure, looking for quantitative explanation)
3. Follow up question -> were mainly regarding this (why this figure)
4. Few question like why Mastercard, like opportunity cost for candidate for studying pgdba

Few HR question:

1. Suppose at mastercard you are the youngest member in team, one disadvantage is that you are asked to do other people are not unable to do then how would handle this situation

Interview Round 2:

Main question were based on case study which was asked in pre interview like

1. what was your approach for case study.

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2. Asked about different aspects of case study like the qualitative and quantitative aspect, what kind of data were helpful for case study, different features that would be helpful, what if some features were not available

Interview Round 3:

Case study -

A bank has 10 Cr customers. It has introduced a new credit card and expecting 10% of its customers to use it. But till now only 1 lakh customers have joined.

How will you help the bank using Mastercard data?

What all KPIs will you look for?

General: Imp - participate in case study round, knowing that whatever solution u r giving question will follow up

16. Yogesh Pal

(B.Tech in CSE from IET Lucknow, graduated in 2020. Worked for 12 months as an Assistant Systems Engineer at TCS)

Round 1:

Technical round, it involved a case study: to design a recommendation engine for a sports streaming platform. Questions were based on GAN architecture, my projects, and statistics.

Knowing what you have written in your CV is a must. Try to cover all the concepts on which people in your batch have done projects. I never studied recommendation systems, didn't do any project on RSs and obviously it was not mentioned in my CV. Still, the interviewer expected me to know everything my batchmates know. So, if possible try to have a glance at the basics of everything.

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17. Srivatsa Kulkarni

(BE in Electronics and Instrumentation Engineering from BITS Pilani, graduated in 2019. Worked for 3 years as a Project Engineer at Wipro. Got to work on basic data related projects, involving things like EDA and outlier detection, for a year)

Shortlisting: Although I had put AI Garage higher in my preference, I was shortlisted for DnS probably because of my below 7 undergrad CGPA. I did submit the DnS Case Study. It was a customer segmentation/analytics case study (You had to assume some data and propose a solution; the focus was mostly on the business part). Approach: I looked at spending patterns of customers, based on banks' credit card data - high, medium, and low spenders. High spenders could be ones looking for convenience and minimum effort and use their cards on lots of different things (to whom we would want to provide features such as no upper limits, tailored more towards no interest loan). Medium spenders have a good credit limit but generally use it only when there are big offers or promotions (some behavioural strategy to nudge them towards payment).

Round 1:

No technical question was asked in this round (Interviewer seemed to have little knowledge of proper analytics)

1. Asked for a walkthrough of my resume (I started with undergrad and work experience and then I moved onto my projects. I highlighted my regression project and asked if she would want me to elaborate on this. She asked assumptions of Linear regression.

2. Case: Platinum card it's been 2 years the credit card spending is low and customer base has not increased. How do you make it profitable? (I literally pitched the same solution that I had for the DnS case competition, since this was also a customer segmentation case. She was okay with it. She was more interested in the customer segmentation approach, not data science aspects)

3. Why MasterCard?

4. How comfortable are you with a client facing role?

You can go through the annual reports of companies to get an idea about their business (Report contains a description in the beginning).

Round 2:

1. Tell me about ONDC (from the DnS case. I referred to Amazon in my answer)

2. How will the advent of ONDC impact Amazon in India? (I said Sellers will probably start moving to ONDC, but the interviewer expected more. I got stuck here and I feel this was the deciding factor)

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In one single Amazon transaction what all ways can Amazon make money? Commission from sellers, from Banks for credit card promotions etc

3. How MasterCard can help the government (I wasn't confident in my answers here)?

4. Any other questions?

Overall there were hardly any technical questions. Develop your business acumen by understanding how businesses are able to generate revenue etc.

18. Aishwary Srivastava [Joined]

DS- Case study competition, 18+6 people shortlisted(internal shortlist also), How would you present, video of you explaining slides(5m)

AI garage – 25 (cg+relevant work ex + software dev + profile based)

Interview Round 1:

1. Tell Me About Yourself
2. What do you know about Mastercard?
3. How Mastercard makes money?
4. Case1: credit card to choose top 1% customers for offering a new scheme which gives reward worth 2000₹ Answer:
We have data of user, for each user we have profiling, segregate diff user according to spending habit, interview helped

Case 2:- new phone launch similar to iphone, which consumers to target for discount Answer: You have select the consumer for credit card, find out phone user those spending habit like iphone, not satisfied, interviewer changed the question

1. You've put AIG as 1st preference and D&S as 2nd. Could you explain?
2. Where do you see yourself in your career in next 2-3 years?
3. Do 28*56 without pen and paper
4. Tell 2 skills you'd like to focus in next 1 year.
5. Given the knowledge on ML you have; do you think you're ready for DL?

Interview Round 2:-

1. What do you know abt mastercard? How do we make money?
2. Full form of MDR?

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3. Who's going to win Amethi?
 4. How many times has your name been mispronounced?
 5. Given you've done so many ML heavy things in coursework, what if the job involves Tableau, SQL and not much ML?
 6. Location preference
 7. Any questions for us?
-

Mastercard AI Garage

1. Manvinder Chahar

The interview was kind of rapid fire. The questions asked were:

- a. Basic intro,
- b. Question on Regression Project \ Business Applicability of project \ Difference between Ridge & lasso Regression
- c. Random Forest
- d. Difference Bagging & boosting
- e. About Time Series Project
- f. What is Stationarity in TS
- g. What are Conditions of Stationarity?
- h. Almost Every Letter in CV was asked.
- i. What is ARIMA
- j. In Classification, How Decision tree Splitting happens.
- k. Difference between Bagging Xclassifier & Random Forest Classifier

Deep Questions were asked from these topics

- a. Rank Of A Matrix
- b. How can it be explained more? Laymen Terms? Explained via Lower Dimensions in [Interviewer was Not much satisfied]
- c. Explain Vectors?
- d. What is BERT?
- e. What is the structure of Word2Vec?
- f. Difference between word2 vec and glove?
- g. How to reduce overall time complexity of neural network architecture?
- h. Without using Softmax operation
- i. Basic ques about CNN project [See Project]
- j. Experience of any Kaggle Competitions ? If Participated
- k. Talked about Assignment on kaggle: Pics to Pics Modelling of Image Reconstruction

Question on Hobbies: Geopolitical

- a. Summarise what happened in the world: BRIC summit: some countries were added?
-

Mastercard AI Garage

2. Zara Rahman [Joined]

Selection Process involved: There was a CV shortlisting followed by two interviews. Basis of CV shortlisting was unclear, but most of the applied CVs got shortlisted

INTERVIEW ROUND 1 :

1) Was this a technical or a business round?

Technical

2) Did this round involve any case study (if yes, kindly explain the problem and your approach)?

No

3) Kindly list down all the questions asked and your responses to them. Mention the source of the questions too.

- a. Cases where random forest performs better than xgboost;
- b. what is the mathematical expression for feature importance with random forest;
- c. Role of generator and discriminator in GAN which their mathematical expression (loss fn); what is logit;
- d. What is cross attention;
- e. What is multi modal transformer;
- f. Why pix2pix worked best in your project;
- g. Did you try autoencoder first; if yes, compare their results;
- h. When is stemming chosen over lemmatization;
- i. why do you think bert performs better than other algorithms;
- j. Explain how word2vec works with dimensions;
- k. Advantages of glove over word2vec;
- l. What are the parameters and hyperparameters in an encoder-decoder model;
- m. What do you mean by latent space in autoencoder;
- n. What is probability distribution;
- o. How is the concept probability distribution used in variational autoencoder;
- p. What is bayes theorem;
- q. Simple question on bayes theorem.

INTERVIEW ROUND 2 :

1. This was a technical round
2. Questions asked are:
 - a) What is lasso and ridge regression;

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- b) which is better for feature selection and why.
 - c) why is ridge regression called so.
 - d) when do we use elastic net regression.
 - e) what do you understand by bias variance trade-off;
 - f) why is there a trade-off between them;
 - g) how to check multicollinearity,
 - h) what to do if you find it, why is it a problem.
 - i) Explain principal component regression.
-

3. Writabrata Bhattacharya

INTERVIEW ROUND 1

The interview lasted for 30-35 minutes. Following questions were asked:

- a. Introduction
- b. Something which is different in PGDBA from CMI
- c. What are loss functions? What are the different kinds of loss function? Explain below ones
 - i. Cross Entropy Loss Function? Binary Classification & Multiclass classifications setting?
- d. What is SVD? - Learn from CDS - DL Course by Mitesh Khapra Youtube - Website
- e. What do matrices represent in SVD ?
- f. Project on Content based recomm system - Que asked on Collaborative Recommender System?
- g. What is Collaborative Recommender System?
- h. Explain Item - Item Collaborative Recomm System?
- i. What is TF - IDF?
- j. Design a recommender system for Blinkit?
- k. Do you want to work on problems given by organisation (like dashboarding) or will work on problems by your choice?

Overall Advice:

Preparation is same for all company - Follow tips mentioned in my BCG's experience

Recommender System - Learn all types if any one is in cv -

Eigenvalue decomposition

CDS is important. Linear Algebra is imp

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ML & DL & SPA are important.

HR Ques: Always mention: whatever company gives, I will do that

4. Raahul N [Joined]

INTERVIEW EXPERIENCE ROUND 1

- a. So tell me about the history of language models
- b. How does attention work?
- c. What is one advantage of attention based architecture over LSTMs?
- d. How is chatGPT different from a normal transformer?
- e. Can you tell your rank in the class?
- f. What is KS statistic?
- g. What is the rank of a matrix?
- h. What is type-1 error? Show it in a confusion matrix
- i. What are metrics used for classification tasks?
- j. Give me an example where you focus only on precision
- k. I see your hobbies include dancing, will you form a dance group in the office (if you intern with us)?

INTERVIEW EXPERIENCE ROUND 2

- a. What questions were you asked in round 1?
 - b. Case on fraud detection using data of online transactions (unbalanced).
 - c. What is regularisation? Types and which one will you use where?
 - d. Can L1 be seen as a feature selection method? Why?
 - e. Do you have any questions?
-

5. Lagensh TS :

A case study competition was conducted by Mastercard before the interviews. Then the shortlisting was done based on the participation/non-participation in the competition and resume.

Interview Round 1:

- 1) Was this a technical or a business round? - This was a technical round.

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- 2) Question asked - Tell me about yourself. What is multicollinearity? What is regularization? Is regularization meant to be used for removing multicollinearity and feature selection? What is a normal distribution? What is a covariance matrix? What is the difference between correlation and covariance?
- 4) Difficulty of round - This round was easy. 4) Difficulty of round - This round was easy.
-

6. Bhavaraju Nikhil Kanth Goutham

It was a Technical round

Case Studies were Not asked

Following questions were asked in the interview

Introduce Yourself,

Questions on projects (different terminology in cv):

- Bias-Variance tradeoff,
- Boosting vs bagging,
- Does boosting reduce the variance

Overall Interview Experience:

- Interview was good
 - I could have been a little more active, like starting to discuss projects on my own, how their company research is working well in this domain etc.
-

7. Nikhil garg:

Shortlist: May be on B.Tech grades

Interview Round:

Machine learning concepts svm , SVD and roc curve, activation functions, how roc curve helps.

General Advice: Very technical learn machine learning at good depth.

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8. Avinash Kumar

Interview Round 1:

1. It was purely technical round and based on CV. Asked about project and follow question. As
 2. Asked about term in CV.
 3. SARIMA, ARIMA diff, Transfer learning
 4. Data augmentation, PCA, why PCA used , how principle component is decided, plot ,no of components, k-means clustering
 5. Max IR people selected.
-

9 . Himanshu Devendra Aswal [Joined]

The pre-PPT case study is mandatory for both AI and DNS candidates, not solely for DNS roles. The submission requirement applies to both groups. Evaluation of the PPT is a black-box process. AI garage had 2 rounds (Gentleman - Pgdba batch 1, Lady - Data science head) - Crisscross

Interview Round 1: Lady (very friendly)

1. Your introduction
2. What is regularization, and why do we use it?
3. About previous work experience
4. From the project, (Linear regression) - How did you do feature selection using L1 and L2 regularization?
5. State 5 assumptions of Linear regression and explain them

HR questions:

1. Why Mastercard?
2. Questions evaluating general interest.
3. No DL questions, No coding, No SQL were asked

Interview Round 2 - Gentleman(Pure technical) - 20 mins

1. What is GAN?
2. What is bootstrapping? More about Random forest
3. Sampling size of random forest, how do you apply bootstrapping on Randomforest?
4. Pytorch vs TensorFlow (no follow-up question)

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5. Uses of Graph neural network and graphs (Himanshu pitched Graphs)
6. Shapley value - Explain SHAP in detail (from resume)

HR

1. What are your hobbies?

General tips: He used YT channels to learn everything - Statsquest, Krish Naik for DL, Ritvik maths. Be good in talking skills, and don't sound mugged up in interviews.

10. Avinash

Interview Round 1:

1. Revolved around CV
2. Bag of words, Tf-idf
3. Bagging, boosting, decision tree.
4. Bias variance trade-off

Interview Round 2:

1. Problem statement of case competition mentioned in the resume.
 2. What is P value
 3. How's pruning done in XG boost
-

11. Kriti Jain

Interview Round 1:

1. Intro - questions on work experience
2. Graph Project - How was the graph made, explain the project in brief. How did the pagerank algorithm work here?
3. How was the algorithm better than the traditional methods
4. What is BERT and LSTM? What is their architecture?
5. What is TF-IDF, given its formula and why it was designed.
6. What is KS-Statistic, how did you use it in the mentioned project

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7. What do you understand by regularisation, name some regularisation techniques
How will you derive the ridge regression loss function and what will it signify

Interview Round 2:

1. Walk through your resume
 2. What is pagerank, give its formula. How was multi-hop used?
 3. What is random walk
 4. What is markov chain
 5. Given a graph except pagerank what all features could be made
 6. What do you mean by Binomial distribution
 7. What do you mean by p-value
 8. Question about hobbies
-

12. Monal Goel

(B.Tech in Chemical Engineering from IIT Kanpur, graduated in 2019. Worked for 2 years as Process Engineer [in Operations and Research & Development, majorly on drug development] at Dr. Reddy's Laboratories)

She had submitted the DnS **case study**: It was on customer segmentation. There was a platform for which a reward system had to be developed. Credit card would give rewards enabling people to purchase from that platform. Segmentation had to be done to identify which customer would be willing to buy that credit card. We had to research and see which analytics tools could be used. Main idea was how we think for a case study. How this business problem could be converted into a data science problem. We didn't have to do any coding but had to provide a data science approach to this problem.

Her approach: I searched about the platform, developed customer personas (on what criteria their buying behaviour depends) and based on that I devised the methodology to target them. I did this for both buyers and sellers on the platform. Also decided on the ML techniques, which features to select (based on demographics, buying history, platform related details, what are they looking for, how we can contact them, what kind of nudges would push them towards buying a credit card : I developed features for all of these)

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CV shortlisting was done (most probably)

Round 1:

Lasted 20-30 minutes. (One interviewer)

It was a technical round

Case study: Credit card frauds need to be detected. Company has merchants and customers. They wanted to develop a graph-based approach to solve this problem. I told them what the features of the graph could be - we have nodes & edges, so we could develop a graph structure and how. Then I moved to how we could use GNN architecture. Then they asked what are the supervised or unsupervised techniques in graphs. I mentioned a few of them. Then we moved forward to community detection. They were checking my approach. They didn't ask any proper business questions.

Question from a case competition (CV based) : There was a food chain aggregator, and we had to help it add on cuisines & customers based on profiles we developed. What K-means, what's silhouette score, in which cases k-means fails, what the pre-processing case for k-means, DB-scan?

Questions from projects: They asked about image processing, CNN structure, what's VGG-16, what's Resnet, what's the benefit for CNN over MLP, why do we do pooling, what are its key advantages? I drew the CNN structure and then they asked questions related to it. They asked about receptive fields.

Random Questions : Eigenvalues & vectors, rank of a matrix, SVD and PCA. They wanted to see if I knew these concepts and given a matrix, how could I derive these. I could answer these questions.

They asked some questions around my hobbies (which was dancing)

Round 2:

~20 minutes, 1 interviewer

It was a technical round. No case study was asked:

Project related questions : What's SMOTE, how to use SMOTE, whether and how it can be used when we have categorical data? What's regression, what's lasso and ridge, can both be used and in

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which cases one performs better than the other (I had mentioned both SMOTE and Regression in my projects. They picked the keywords from there and then asked questions on those topics).

Undergrad questions : What did you like about IIT Kanpur?

Some general questions from my work experience

No business question was asked in this round either

13. Anant Yadav

(B.Tech in Mechanical Engineering from IIT Kharagpur, graduated in 2019. Worked for 37 months at Tata Steel, 12 Months as Mechanical Maintenance Manager and 25 months in Project Management)

He had submitted the Dns **Case Study**: Building a recommendation engine for a mobile application of ABC Bank, which will suggest to users different kinds of services provided by the bank. No dataset was given, and we were supposed to make a presentation explaining an approach

My idea was to use two kinds of dataset : transaction data & customer profile data. My approach was to cluster customer profiles for a particular region, and then I would build a separate recommendation engine for each of these segments. I relied on K-means & DBScan for this clustering. Regarding customer profile, my chosen variables were : bank balance, credit score, age, monthly income, etc. For the transactions dataset, I relied on RFM (total transaction amount, monthly frequency, average amount of the transactions, etc. Once I had these customer segments, I proceeded to build the recommendation engine, which was built using RFM analysis, which I did for each user segment and product/service offered by the bank. Based on this information I found relative scores. I then adopted a collaborative filtering approach using these data ; customer segments, product/service and corresponding score I had calculated (scaled from 1-10). On the technical side, I build the recommendation engine using word-to-vec architecture, along with negative sampling and skip-gram model.

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Financial knowledge wasn't required as such for this project. I spent 8-10 hours on this project. We also had to make a 5 minute video.

Round 1:

Interview lasted ~30 minutes. Given by the director of AI Garage.

I had mentioned 2 terms - time based maintenance & condition based maintenance - in my work experience section. She asked what they meant. Then she moved to my time series project. She asked what this project was all about, why did I choose this (S&P 500 index forecasting project). I said investment opportunities that index funds offer. She asked for index fund details. Then she asked what's the SARIMAX methodology, what's ADF test, what's stationarity (I had mentioned this word in my description), what's strong and weak stationarity, what are ACF & PACF functions, Ljung-Box test, how did I make an optimal model selection (I used the BIC criteria). She asked "Why BIC?" I answered that there's a bias-variance tradeoff, BIC prefers a simpler model hence there's a higher bias and since I have a small number of observations, I didn't reduce my variance and went for bias.

Then she moved to my 2nd project that was regression analysis on customer expenses prediction. She asked "What's meant by OLS; what are lasso, ridge and penalty in linear regression; what did I choose Lasso & Ridge (since my dataset had interaction among the predictor variables, I decided to expand my feature space using polynomial feature expansion, and I did that using degree 2 polynomial feature expansion. For this, I could either use all features I had and then reduce variance ridge, or perform automatic feature selection using Lasso). Then she asked "What's backward feature selection?"

Then she moved to my Deep Learning CNN project on assembly line scheduler via image classification. She asked: "What's this project all about? What was my basic methodology?" Then she asked me to draw the structure of LSTM

Towards the end she moved to my 5th project that was Search Classification (binary classification). I had used Active Learning here, and she asked what it was, why did I choose active learning only in classification and not regression? (Active learning is based on an uncertainty principle, which can only be described for a classification problem, not a regression problem) I had used RF, Adaboost & XGBoost algorithms for this. She asked why only these and not others (I said I had used logistic regression, SVM and dec trees as well but those 3 were the top 3 performing). She asked how I

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determined the performance (I said F-1 score was my criteria for selection. Active learning is an iterative process, I observed the F-1 score trend for training dataset over number of iterations. RF & XGBoost performed best here)

~50-60% people cleared this round

Round 2:

~30-35 minutes. Given by an alumnus of Batch 1.

This round was a bit more spicy. The 1st question he asked was "Tell me about an ML/DL algorithm that none of your batchmates know?" (I described how I used word2Vec arch to generate graph embeddings for a graph with negative sampling using skip-gram model, which I had used in my recommendation system project). He asked what my recommendation system project was about; what's collaborative filtering, why did I use collaborative filtering, what's meant my nearest neighbour graph, how do I construct it, What's the random walk that I've used here, the 2 models of Word2Vec architecture, what were my performance metrics (I had used intra-cluster similarity). He asked for an alternative approach/metric (I said OLS, he was okay with this). He asked "What's meant by eigenvalues and eigenvectors (in this context, he wanted to know a special property that when any matrix is multiplied by a vector, it performs a linear transformation involving shift, translation and rotation. Eigenvalues are those directions along which there's only scaling, no rotation). He asked "What's meant by PCA and SVD?" (PCA was mentioned in my CV and SVD is also kinda used in PCA, but this question was being asked to everyone by this panelist). He asked "What are the mathematical formulations for SVD & Eigenvalue decomposition as a special case of SVD; what's the relationship between SVD and PCA?" (If you look at the formula, SVD used a rectangular matrix. In such cases, you compute $X^T X$ and XX^T , and you compute the correspond eigenvalues and eigenvectors for both; it so turns out that eigenvalues for both are same, theoretically. He was looking for the $\text{UsigmaV}(t)$ formulation basically; in sklearn, after standard scaling the entire data, the mean vector becomes 0, so $X^T X$ represents nothing but matrix for covariance). He asked how my B.tech experience at Kgp was different from PGDBA experience.

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14. Anurag Shukla

(B.Tech from Indian Institute of Space Science and Technology, graduated in 2019. Worked for 3 years as a Scientist (core electronics) at ISRO)

He did not submit the DnS Case Study

Round 1:

Technical Round, ~25 minutes. All questions from CV.

She started with my regression project and asked similar questions (to BCG): Ridge, Lasso, etc. I gave similar answers

Then she moved to my NLP project. Exactly the same questions as in case of BCG.

In my CV, I had a couple of projects on CNN, plus my published paper was also on CNN. She asked not much work was being done in image segment in MasterCard. How would I contribute? I answered that the basic idea of deep learning is same, whether we are working with CNN, LSTM, or anything else. I bring that basic idea to the table. Hence, I can work with any type of neural network architecture. I gave examples of other projects that I had on other deep learning methods.

Round 2:

Questions from ML techniques (mostly those mentioned in my CV). What's XGBoost, why is it better? What's Word2Vec architecture? He asked me to draw Word2Vec architecture on paper (not mentioned in my CV). I drew it, upon which he was pointing at different parts of the drawing and asking for basic explanations therein. What's SVM? How is classification different from regression in SVM? What are kernels in SVM? How do kernels help with SVM classification? On SVM, I answered "Suppose we have two classes, A & B, we can draw a line between them to separate them. SVM draws two parallel vectors instead, and it tries to ensure that the 2 classes are on two sides of this vector and distance between these 2 vectors (effectively distance between these 2 classes) is maximized; In classification, values on either side of these vectors are the classes. In regression however, having found these support vectors, we find the hyperplane that passes in between these 2 support vectors, and the value on this hyperplane gives us regression output; Kernels are used to transform data from non-linear high-dimensional set to an easy-to-interpret from. Different types of

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kernels are linear, exponential, RBF." He asked "What's RBF?" I said "It's Radial Basis Function. It scales our data into infinite dimensions. It's the most used kernel, as we need not specify any information prior to applying this kernel, unlike in other kernels where we need to specify the dimensions, which may not be easily known always. Hence, RBF is preferred". He asked "Is RBF used anywhere else?" I answered : "Kernel is a concept of Mathematics used to transform data from an n-dimensional vector space to an m-dimensional vector space." Lastly, he asked, "How good are you in programming? How would you rate yourself on a 0-5 scale?" I answered that I am good with the logic of a code, so given the syntax, I can put anything into code. So, I rated myself 4/5.

This interviewer had a very fixed set of questions in mind (XGBoost, Word2Vec, SVM; regardless of CV), and was expecting a fixed set of answers. There was no guidance in any question in this round. He was expecting tech-bookish answers.

15. Ayushman Yadav

(B.Tech from Mechanical Engineering from IIT Guwahati, graduated in 2020. Worked for 8 months as a Graduate Engineering Trainee at Reliance Jamnagar)

Round 1:

Technical Round (~15 minutes)

It started with my introduction, where I mentioned few of my projects. However, I did not mention my recommender system project in my introduction, but they asked questions from it: "What's a recommender system, what different types of recommender system did you use, what was its ultimate objective?"

Then they asked questions from my regression project. I explained my project. Then they asked "What's Regularisation? What is it used for? How is regularisation applied in the context of regression? Why is Lasso better than Ridge?" (I explained that regularisation is applied by adding one more term in the objective function which also have to be minimized; Lasso can be used for variable selection as few of the betas will go exactly to zero)

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It was sort of a rapid fire round. They were firing questions as soon as I mentioned any relevant term in my answer.

16. Aravintha Kannan

(B.Tech in Mechanical Engineering from NIT Trichy, graduated in 2019. Worked for 18 months as Engineer - Product Development at Mahindra Research Valley)

Round 1:

Technical Round. No Case Study was asked.

1. What is Multicollinearity? My answer: In a linear regression framework, when 2 independent variables are correlated with each other, you can't assess independently their impacts.
2. Which is better- Ridge or Lasso Regression and give examples when you would which : Didn't answer properly
3. Explain XGBoost. Write down its loss function equation : Just gave a broad overview, couldn't write loss function. Be ready with mathematical part of any algo you mention on resume
4. What is CatBoost ? : Extension of XGBoost. Gave was a broad overview

All questions were from resume.

You have to be technically very clear with an in depth understanding of concepts, can't beat around the bush. Deep Learning projects with hands on Tensorflow/ PyTorch will help you in the interviews.

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1. Rahul N

INTERVIEW Round 1:

- a. Walk us through your resume.
 - b. Tell us about your time-series project.
 - c. What are the problems you face when you predict too much into the future?
 - d. Mentioned MAPE would be same for errors of first and last month of prediction
 - e. Did you decompose your time-series data & What kind of seasonality was present in your data (additive or multiplicative)?
 - f. How do you find when it is multiplicative/additive?
 - g. What exogenous variables can you think of to use here?
 - h. What are the problems in using Temperature as an exogenous variable?
 - i. If there is a trend in the temperature, how do you find it?
 - j. Did you plot the distribution of y? How did log transform work on the right skewness?
 - k. What is Box-Cox transformation? Why did you use it here? Is it only for right skewed data?
 - l. In your insurance cost prediction project, what were the variables? What other variables can you use & where can you get it from?
-

2. Zara Rahman

CASE ROUND 1:

1) What was the case competition about?

Response: Time series case to predict mobility index

2) How did you approach it? How much time did you have?

Response: Used few exogenous variables, tried multiple approaches like arima sarima fbprophet xgboost random forest

3) On what parameters was your code evaluated?

Response: RMSE score, good coding practice, exogenous variables

4) How difficult/competitive was this round?

Response: on a scale of 1-10, 7 was the difficulty

5) Was there any other shortlisting criteria:

Response: Resume Based Shortlisting also happened

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INTERVIEW ROUND 1:

- 1) It was a technical round
- 2) no case studies were asked
- 3) Below is a list of questions asked:
 - a. What is feature importance
 - b. What is bias variance trade-off
 - c. What is probability distribution?
 - d. 3 numerical questions on probability
 - e. When do we use ridge regression?
 - f. Two puzzles (geeks for geeks)
 - g. If you are given a dataset with many rows >one million, how will you do clustering on such data.
 - h. Is it okay to have multicollinearity in your data?
 - i. There were other questions of similar level (can't recall)

Overall experience of this round was: easy.

INTERVIEW ROUND 2:

- 1) It was a technical round
- 2) I was asked two cases:
 - a. Case1 - You have a lot of hospital data; how will you make use of them as an investment firm to invest and earn.
 - b. Case2 - you have the footfall of a restaurant for the last 5 years. At this point covid happened and there is lockdown, so footfall is zero for the next 1 year. Now how will you predict the footfall for the first day when lockdown is removed and for the first week?

INTERVIEW ROUND 3:

- 1) It included technical / non-technical questions.
- 2) Case Study:
 - a. You want to open a port in India, what factors will you consider?
- 3) Other questions asked to me are:
 - a. SQL queries (partition based),
 - b. Python questions,
 - c. what are the metrics involved in regression and classification, which is better in what case, what is harmonic mean and when is it used a metric

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- d. Why is bias variance tradeoff important ?
-

3. Manvinder Chahar

Initially, there was a Case Competition: It was a Time Series problem to predict Mobility Index. Compare it to a benchmark and see what is average movement?

Tasks I performed were identification of Exogenous variables, Regression Based TS Model, ARIMA. 3 Teams won & Shortlisted for Presentation Round. Apart from this, there was normal resume based shortlisting. I made some Mistakes in Model creation: Took Exogenous variable of same day for which prediction was made. There must be a lag between predicting time for exogenous variables.

INTERVIEW ROUND 1

- a. Introduce Yourself
- b. Questions on Time Series Project
- c. What is ARIMA, ACF?
- d. If I increase the number of trees in random forest, how will it affect Overfitting?
- e. Bias & tradeoff in Machine Learning

Other Questions asked to me in this round are:

- a. What is application of Flight price prediction
- b. If you have to define price modelling for an airline company, what all factors should be considered?
 - i. How fast seats are filling
 - ii. How many seats are left
 - iii. Is it Holiday
 - iv. Price of same flight in last year, last quarter, same day of last week
 - v. Take care of seasonality
 - vi. Price competitors are changing
- c. In covid scenario, as an investor, how will you identify which airline company to invest in?
 - i. Counter question: Huge loan from bank but has own planes vs company who has taken planes on lease? Which one will you consider?
 - ii. I answered Planes on lease.. since negotiation can be done for lease but not for loan.

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- d. Basic Ques on ML:
 - i. Diff between Random Forest & XGBoost?
 - ii. Lasso & Ridge
 - e. Ques on Macroeconomics
 - i. Explain Inflation
 - ii. How will you measure: CPI / WPI?
 - iii. What is the law of demand?
 - iv. What is repo & Reverse Repo Rate?
 - v. How does RBI operate in such a scenario?
 - vi. Technical Definition of Recession?
 - vii. Different Type of Inflation: Demand Pull & Supply Side
 - viii. Yield Curve- How price changes with maturity of bond?
 - f. Interviewer asked if Any Ques for me:
 - i. What does the 1st month at Deshaw look like?
-

4. Arpan Singh

In my view, the company prefers Analytical Background.

Interview Discussion Focuses on Regression & Relies on Basic Things Like Linear Regressions: Goes into Depth of Easy Topics in Pre-Round. Time Series is very important for Deshaw.

Deshaw Organises a competition every year: If in Top 3, You can skip the Pre-round

Interview Round 1: (Lasted ~35 Minutes)

- a. Discussion on any Project: Like Time Series, Regression is usually done
- b. Assumptions of Regression. What if they Fail?
- c. Why do we need to test for normality of Residuals?
- d. What is QQ Test - Came from Resume
- e. What is Multicollinearity? How do we check it? What if it is present in data, what to do with the model? Basic Questions. What is the formula of VIF?
- f. Why do we need to standardise our variables before applying ridge-regression?

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- g. Classification Metric: Recall, Precision, Accuracy?
- h. Which is used in which situation & How to Judge? Give Business Use cases where Precision is preferred over recall?

Tip: All these can be covered from ISLR - Deshaw's Tech is covered from ISLR & Inference - from previous year notes.

- i. Hypothesis Testing: How Do You Check Coin is Biased or not?
- j. Type 1 error. Calculate Power & How do you increase the power of test? Show complete process of it - This was very important: Everything was judged using this question - Inference Notes
- k. Puzzle - There are 100 Matchsticks-Two Player games- One each person can pick 1,2,3 or 4 matchsticks on each turn. & whoever picks the last one loses. So tell the strategy to win & Who should play first.

INTERVIEW ROUND 2:

- a. Introduce, Talk About Previous Work-ex.
- b. Already had Work experience in Currency & Investments
 - i. How Prices are regulated in different exchanges?
 - ii. Ques related to Project in Work Experience.
 - iii. During covid, the currency of Thailand was reduced, What could be the possible reasons?
- c. Gave a case: Suppose you have to Open an airport in India; Decide which city?
 - i. How to approach a problem,
 - ii. Open Ended case; Thinking Ability
- d. Asked About Deshaw's competition
- e. Asked about time-series problem
- f. Asked Tech Questions to those who were not part of Competition's final round

INTERVIEW ROUND 3:

- a. Walk Through Resume; Work Experience & What work was done;
- b. I was told to choose one project from my resume & questions were related to business on it; I selected a regression project.
- c. Insurance cost prediction
 - i. Why premium collection is important for company,
 - ii. What variables can be taken to dataset for exhaustive understanding

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- iii. Lot of cross - questioning happened here
 - d. There was one puzzle in second round
-

5. Priyanshu

Interview Round 1:

There were 2 People:

Started with Discussion on Work Ex of Axis bank

- a. Impact Created by work done at Axis Bank ?
- b. Cross Questions on Everything Said

Next Ques were asked on NLP Project:

- a. What is this project all about, business implication; how did you go through it?
- b. Very Important: Features that you took. Why?
 - i. Any Review That was very large or lot of exclamation mark - is an extreme review.
 - ii. No. of words,
 - iii. No. of exclamation marks.
 - iv. Above features were fed in the model
- c. Mentioned: Tableau.
 - i. How comfortable with Data Visualisation?
 - ii. How comfortable with Tableau? What is level of details in Tableau?
- d. What is the difference between Tuple & List? Use-cases

Interview Round 2:

- a. Any Specific Feedback for us from previous round:
- b. Was there something you should have spoken in the first round?
 - i. Talked about Myers Brigg Algo.
 - ii. Discussion initiated on M-B Algo
- c. Gave Case on Hiring: A firm has to hire at a new location: What parameters would be considered ? Business Case - Went Well
- d. Another case on IT company trying to expand in some new location - Approached like a Data Science Case and Didn't Go Well

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6. Writabrata Bhattacharya

We were given a Forecasting Problem- Time Series: where Model Accuracy was tested; Python script works or not and How Good is PPT were parameters considered.

Caes study was related to: "Forecast the mobility index of Euro during Covid Year; Historical data from Feb'2020"

Solutions were Presented. Discussions occurred around below points:

- a. Cross-validation process: Why expanding window & why not sliding window process was selected?
- b. Why were there spikes in ACF and PACF plots after seasonal decomposition?
- c. Talked about - Exogenous variables. Why were they chosen and what other could be taken?
- d. Why is our approach better than Sarimax and how is it better than your model?
- e. Augmented Dickey - Fuller Test?
- f. Ljung box Test ?
- g. What will happen if there is a Black Swan Event?
- h. Good Understanding of What Covariance; Auto-correlation, correlation are?

Interview Round 1:

Lasted for around ~50 minutes

Following questions were asked in this round

- a. Give your Introduction: Brief
- b. Which Project do you want to talk about? Time Series;
- c. Taxi Demand Forecasting in the USA?
- d. What models were used?
- e. What methodology to solve the problem?
- f. What kind of data: Time Period
- g. Time series decomposition;
- h. Question on whether data is stationary or not; Identify seasonality;
- i. Carry out the trend and seasonal differences?
- j. ACF & PACF plots
- k. Identify the order of your time series model?.
- l. Look at forecast, look at residuals, look Ljung box test for residuals;

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- m. Calculate the MAPE (Mean Absolute Percentage Error)

Next, they gave me a case-

- a. Taxi Service Provider - How will you forecast the taxi demand near the airport- assume any - move from local modelling to global - any airport.

I. Identify the main three to five features for the same.

Provided hints for selection of features;

What are the Exogenous features used & which are missing

INTERVIEW ROUND 2:

- a. Asked for Project of your choice but later themselves chose Classification Project:
- b. Explain the project. Why choose this model? Why logistic model a linear model & Why XG Boost is better than logistic regression model?
- c. What features were there in the dataset? What other features could have been looked at?

Next, I was given a Case Study with below problem statement:

Some Hotel in Gurgaon: Deshaw provided a loan to them. Deshaw wants to know how much revenue is the hotel going to generate over the next two years?

Those two years are covid years. - Kartik & Rohan also had similar case

- a. Different factors you see for their revenue?

Then there were questions on:

- a. What is hypothesis testing? What is p-value?
- b. What is precision & Recall & ROC Curve?
- c. What is F1 score the harmonic mean of precision & Recall?
- d. Do you want to join BCG or DeShaw?

Advice:

- a. Go through all the past interview questions
- b. Puzzles from Geeks for geeks;
- c. Probability questions: 50 probability questions;
- d. Very Imp: Time Series; Regression; Classification - very thoroughly
 - i. Rob Hyndman - Book on Forecasting Principles;
 - ii. Stat510 by Penn State;
 - iii. RTSM Course in KGP;
 - iv. Couple of slides: SAAPM in IIM-Cal

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7. Mayank Kale

INTERVIEW ROUND 1:

Questions related to

- a. Mostly basic rtsm.
 - b. High level finance and economics knowledge
-

8. Ramnath

Applied for DESIS Global Analytics and reporting, shortlisted for the same.

Lasted for 70 mins.

The interview experience was good. They were calm and listening to the answers.

What should have been done better - Should have prepared better for HR round

Interview Round 1:

1. Was it technical round or business round - Technical Round
2. Were any case studied involved - Yes, a couple of case studies.

Case Study 1: A tech company facing high attrition propose an analytics-based solution. They will drop more information & hints as we explain/ask more.

Broke down the problem statement to understand the cause. To see if it's a global/local issue. Explored possible causes.

Case Study 2: Gave me an Excel file containing some employee data. Asked me how I would approach it to improve diversity at the workplace.

Same as previous. Used to excel and explained how a dashboard can be made to understand the possible causal factors like department, managers etc.

3. More questions on me and my resume.
4. Difficulty of interview - They were evaluating the thought process. Not very hard as long as u have read IIM A consulting casebook

Interview Round 2:

1. Technical or business round Non-technical round
 2. No
 3. Why DEShaw? Why GAR team? Why HR analytics?
 4. It was more of a HR round.
-

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9. Emayavaramban A

I gave HR analytics as my preference and was shortlisted based on my resume.

Interview Round 1:

1. It was technical round.
2. There were questions about my resume projects, why do you want to join HR analytics and of course, tell me about yourself.
3. There were some questions related to data visualization as well

Interview Round 2:

1. It was a technical round as well.
 2. I was given a case study also(There has been a case of employee attrition and the company has started to give salary hikes to employees who are giving their notice.
 3. Seeing this, the loyal employees of the organization feel betrayed as their colleagues are getting paid higher, just because they threatened to quit the company.
 4. This has created unrest in workspace.
 5. How would you tackle it). I tried to provide an answer but the panelists doesn't seem to be satisfied with my answer.
 6. They asked me how well versed are you in Power BI/Tableau and asked some questions regarding the same
-

10. Lagnesh :

1. The case competition was about forecasting the daily mobility index of Europe using statistical/machine learning/time-series models.
2. Approximately 1 week time was given for submission. Team size was 2. We approached it as a time series problem with exogenous variables.
3. Presentation was evaluated based on Summary, EDA, Data pre-processing & Feature Engineering, Modelling, Insights.
4. This round's difficulty level was moderate to difficult.

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5. The 3 winning teams from the competition were given opportunities to directly appear for the interview whereas others had to go through shortlisting and pre-interview round.

Interview Round 1:

This was a technical round (pre-interview round). This round didn't involve any case studies.

Questions asked:

1. Introduce yourself.
2. Can you pick any two projects which we can discuss?
3. Explain the dataset in Financial Health Assessment project. Where did you get the data from? How many features were initially present? How did you narrow down the features for statistical analysis? Why didn't you perform the regression on all features and then narrow down the features?
4. What will you do usually when the literature says some features are not related to dependent variable - Will you analyze them using statistical techniques to arrive at a conclusion or will you simply omit them?
5. What will you do if you identify that certain variables are significant but the domain experts don't accept it? What is this phenomenon called? What will you do further?
6. How did you remove multicollinearity using Variance Inflation Factor? What other techniques can be used to address multicollinearity? Are regularization techniques meant for removing multicollinearity? How can regularization techniques remove multicollinearity? Is overfitting caused by multicollinearity? Is overfitting the only effect of multicollinearity? What are the other effects? Will these other effects also be addressed by regularization? If yes, How? How to address multicollinearity if we want to keep all the variables? Apart from regularization, what are the other ways?
7. What is the distribution of the dependent variable (Capital Adequacy Ratio) in the project? Can we apply Linear regression to any dataset irrespective of the distribution of the dependent variable? Will it work on a distribution like Poisson? Is Poisson distribution a discrete distribution or continuous distribution?
8. How do you check the normality of errors/residuals? If normality of errors assumption is satisfied, can we apply linear regression?
9. What are all the other assumptions which need to be satisfied for linear regression?

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10. If given a chance, what would you do now with the dependent variable to improve the performance? What transformation would you apply on the dependent variable?
11. What are all the top significant variables which you identified as influencing CAR (dependent variable). Explain why ratio of total deposits to total liabilities has a negative influence on CAR.
12. Where does the outliers get identified/segregated in a decision tree? Are decision trees robust to outliers? If yes, give 2 reasons for the same.
13. Explain about the class balancing done as part of Fake Review Detection project. Why did you go with under sampling? Why not other techniques? Is there any disadvantage in not going with over sampling? How will you balance between the trade-off between information loss and performance in over sampling and undersampling? In which cases, will over sampling be useful?
14. Explain Word2Vec. How is it generating word embeddings? And how are those embeddings used in the neural network architecture?
15. Explain why and how you used the LSTM in the project. What is vanishing gradient problem? What is the solution to overcome it?
16. When do we use hierarchical clustering and when do we use k-means clustering?
17. What are the features which you used in Medical Expense Prediction? Which feature had the highest significance? Where did you get the dataset from? In real life scenario, from where can the insurance companies get such data? How can the company benefit from this model? What is the expense which you have mentioned here? Assume hypothetically that you have no constraint on collecting any data. Which other features would you like to collect data on and use it in your model to improve the performance? Think and tell about the ways in which this medical expense prediction model can be monetized (don't just constraint yourself to the insurance domain. You can think of any business).
18. What is the difference between error and residual (Think with respect to the observed regression line and population regression line)?
19. Puzzle: You and I and playing a game. There are 100 match sticks. Each person takes 1/2/3/4 matchsticks at a time and then the other person takes turn. The person who takes the last matchstick will lose the game. Tell me a strategy to win this game. Who would you want to start the game?
20. The puzzle was already asked in an interview. So, it must've been in your class's interview questions repository. Why didn't you know that?

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This round was difficult.

11. Himanshu

Interview Round 1:

1. Grilled on AOC ROC curve
2. Draw the curve, and its significance, in detail
3. Can R^2 be negative? Training R^2 cannot but testing can be
4. What are the null hypothesis and alternate hypothesis?
5. Type 1 error, type 2 error
6. H_1 , H_0 are they mutually exclusive and cumulatively exhaustive?
7. Python questions on sorting and aggregation

Round 2:

1. A business case study - nontechnical case study
-

12. Avinash [Joined]

Interview Round 1: two panellists (30-40 mins) - Online

1. Introduction
2. Basic linear regression questions
3. R square
4. How much do you know Python?
5. Suppose you're given stock data, how'd you arrange them, and what Python functions do you use?
6. What recession and inflation, how does RBI monitor inflation and repo rate?
7. Leading and lagging indicator of recession

Round 2:

1. Case study on Hypothesis testing - How'd you test the average annual income in the US is X amount?
2. AOC - ROC curve, when it will be zero?

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3. If you want to construct an airport in India, what parameters would you consider to narrow down the location?
4. Linear regression - Multicollinearity
5. Python questions
6. Do you know advanced SQL? No (they're still fine with it)

Round 2 - 2nd part:

1. Introduction
 2. Pick your favourite project - He picked a Regression project, explained the business problem, and feature selection, and achieved results
 3. Loan prediction project - questions on the project
 4. How do you impute values for missing data?
-

13. Diwakar

Interview Round 1: Case: If you have a GIS data of ships carrying the load, can you predict how many ships travelling in that route and how many tonnes they're carrying?

Answer he gave: If you have GIS data and satellite images, we could do image detection and do something. Based on length and breath, we could estimate the depth of the ship and from there we can find the number of tonnes it is carrying.

14. Kriti Jain [Joined]

INTERVIEW Round 1:

1. Walk through your resume and tell the main highlights
2. On the project chosen - give a brief about the project, objective, process, metrics
What is PCR and how did you choose the components
3. Why did you go for polynomial regression
4. How will you deal with the interaction terms present
5. Why did you go for non-linear models?
6. How would you monetize the project

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7. If any other features were to be included what all features will you include to enhance the performance
8. Can you approach the project by an alternative method
9. What all insights did you get from the project
10. Will choosing R-squared or RMSE make any difference
11. What is difference between Random forest and decision tree
12. What are the hyperparameters of Random Forest
13. How will you perform grid search CV
14. How will you deal with categorical data - Give example of results of one hot encoding and Label encoding
15. Any other ways by which data can be encoded
16. Can we do classification using a regression method
17. What were the metrics that you obtained in your project, the numbers
18. Puzzle -Calculate number of rectangles in a chessboard
19. Case -Given a furniture shop in India, how will you forecast its sales in the new market?

Round 2:

1. Walk through your resume, choose any two projects.
2. Why did you choose OYO after metallurgy?
3. Give brief about regression project, details asked in between.
4. Time-Series project - Tell the details.
5. Given a case on the basis of the project - given a alcohol shop in India, what all exogenous variable will you consider and why
6. Suppose now you have to project number post covid how will you do it.
7. Puzzle: Suppose there are two amounts x and $2x$, you have to choose one - you choose and see that amount and now another person could either have double or half the amount, will you exchange the amount? How will you decide, show your calculations?
8. Why do you want to join Deshaw?

Round 3:

1. Introduction: questions asked in between
2. State Linear Regression assumptions
3. What if errors are correlated, what will happen to the confidence interval. How will it impact coefficients?

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4. Given you have a 5-star hotel, and you are in Dec 2020, you have to calculate revenue and expenditure of the hotel. Under what heads will you break them and what factors will you consider.
 5. Now suppose you have to project bookings for 2021 and 2022. What all factors will you consider, what will be the exogenous variable and how will they impact the different heads that you have broken down.
 6. Why is the manhole round?
 7. Why are pizza boxes rectangular ?
 8. What is recession, what are the factors impacting it.
 9. Hedging - what all stocks will i consider to invest in
 10. Why is F1 score harmonic mean and not geometric mean, give extreme case example
When will decision tree not work
 11. Rate yourself in python - question given on pandas manipulation, given three columns, some product category, dates for 10 years and the quantity sold. Write a code to get maximum quantity sold by products and the date on which it was sold
-

15. Mukul Mundle (B.Tech + M.Tech in Chemical Engineering from IIT Kanpur, graduated in 2018. Worked for 3 years as Assistant Manager - Digital Transformation at Dr. Reddy's Laboratory. Thereafter, worked for 1 year as Business Strategy Manager - Digital Excellence at Jubilant Group)

De'Shaw had launched a **case competition** and I was the winner of the competition, so I was directly shortlisted for DE Shaw's Final Round

Problem Statement: We (group of 2) had been given mobility index (on daily basis, mobility of people coming in an out of shopping malls, theatres, etc.) data for European Market. It was a time series data, and we had to find the trend of mobility index. We had 1 week's time

We used 2 approaches: Basic time series approach where we utilized Sarimax & predictive modelling approach where we utilized certain exogenous variables (which could impact this time series data) and ML models like XGBoost, LightGBM, Gradient Boosting, etc. We had made a really good presentation.

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Round 1:

Tech + Business interview

Case Study: Say, we give you the mobile GPS dataset of people in Kolkata. How will you leverage this dataset for De'Shaw?

Approach: We wanted to try finding clusters in this dataset at different locations, and can be used for various purposes like opening shopping malls, traffic management and control; data could further be sold to government agencies for traffic and infrastructure management or to fleet management companies like Ola/Uber. Interviewers were focused more on how DE Shaw could directly leverage this dataset. Say, DE Shaw wants to invest in a few companies. How could this dataset be monetised further?

Then they moved into my time series project (work experience project). It was about a manufacturing unit and daily electricity demand had to be predicted. We had to utilize exogenous variables like humidity, temperature, total daily volume, etc. I had used the same approach as the De'Shaw Case Comp- Time series (SARIMAX) and ML methods. Their questions: "What were the exogenous variables, why did you choose them, should you have used more exogenous variables, how does the SARIMAX model work, how do XGBoost, LGBM model work?"

They also gave me a puzzle related to blue and red socks and if we had to get a pair of blue and red socks in some combination. It was a probability question. I could solve it (they were looking for the methodology).

Round 2:

In my introduction, I had mentioned that I had interest in microeconomics. They asked "What's the yield curve, what's recession, hyperinflation?" They asked some questions about Indian Monetary Policy. "How are inflation rate and interest rate related to each other."

Case Study: Say, the Indian government wants to open an airport. What should be your approach to check the viability of opening that airport. My approach: Look into the population, demographic dividend, per-capita income of that area, total number of

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passengers travelling by bus/train in the area, is their airport nearby that location, etc. I considered all these approaches and provided a solution. They were content with my answer

They asked a question on python, where we had to utilize groupby and pivot commands (don't exactly remember the question).

They also asked about k-means clustering and KNN ("What's the overfitting & underfitting problem with KNN")

16. Ayushman Yadav ((B.Tech from Mechanical Engineering from IIT Guwahati, graduated in 2020. Worked for 8 months as a Graduate Engineering Trainee at Reliance Jamnagar)

Round 1:

~ 30 minutes

They asked for my introduction, and they asked me why did I join PGDBA. I answered that I had studied statistics subjects in my bachelors and that's where I developed the relevant interest. Then they asked what all statistics related things I had studied in my bachelors, I said basic things like p-value and hypothesis testing. For the next 20 minutes, they asked questions on hypothesis testing (whatever was taught in the Inference course about hypothesis testing [by Amitava sir] were pretty much covered: "there's a coin, you are tossing it, you wanna test whether that coin is fair or not. What will your alternate, null hypo be? What will your test statistic be? What kind of test will you perform (test of proportion)". (I answered all these). Then they gave a question on test of means: "there's a distribution generating some data, but you don't know the mean of that distribution, and you have to estimate the mean using samples" (Don't remember the exact question, but the solution was testing 2 means. I started with null and alternate hypotheses here too).

Then they gave a case study: "Say, you have Domino's data, and last year they had introduced a feature of 1 pizza free per pizza ordered on Saturdays. You have 1 year sales data for this. Using hypothesis testing, you need to check whether this offer is giving significantly higher sales on Saturdays or not." (Again, test of means using average sales on weekdays and average sales on Saturday, and so on.). They asked further: "Using the same data, you wanna see on

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which day of the week the offer will be most effective. How will you do that?" (I couldn't answer this)

17. Anant Yadav [Joined]

(B.Tech in Mechanical Engineering from IIT Kharagpur, graduated in 2019. Worked for 37 months at Tata Steel, 12 Months as Mechanical Maintenance Manager and 25 months in Project Management)

De'Shaw Case Competition: A time series forecasting case competition where we had to forecast the mobility index for a masked out European country. We were free to use any publicly available dataset. We could use any methodology (the simpler and more informative a methodology, the better). There was a leaderboard which showed our model's performance on a validation dataset (rankings refreshed once a day).

We had 3-4 days to prepare and submit (in teams of 2). My team cleared the 1st round. Our performance on a hidden dataset, how well our code was written, the comments put, ppt made, etc. We couldn't win in round 2 due to our poor presentation (top 3 teams were given a direct entry into the final round of De'Shaw)

Round 1:

A 2-panelist interview, both of them were PGDBA seniors. (One of them was the judge in the final round of my cast comp presentation, where he had asked me a lot of time series questions. He didn't ask TS questions in this round to me, but he did so to others). My interview lasted 50-55 minutes

It started with my introduction. Then he directly went to the last project in my CV (which was on customer segmentation) and started asking questions therefrom: "What did you do in this, what kind of dataset did you have, etc." (I had used vehicle insurance policy dataset that contained details about different insurance policies for vehicles of different kinds, corresponding owner names and details about total coverage amount, total premium paid,

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credit score for each customer, vehicle age, total amount claimed, etc. I used K-means, spectral clustering, agglomerative clustering and DBSCAN to identify different clusters, and finally I stuck with K-means and spectral since they were giving almost similar results. I used the evaluation metric of Silhouette score and Davies-Bouldin index as performance metric (also WCSS score in case of k-means)). He asked what other categorical variables were there (but I didn't have any). Interviewer actually had a very good background in customer segmentation. There was a line in my CV about cluster profiling (done after completion of clustering). He asked "What exactly did you do in terms of profiling?" (My approach was to identify the distribution among claims, number of claims, total premium paid; since the best customer for an insurance company is one who pays all the premium and doesn't make any claim). He asked "How can you use this information?" (I found one of the 3 clusters to be very big, since this was a group of customers who had recently bought their vehicles, and had made minimum claims and had paid among the highest premiums. Distribution of vehicle age was also small. And for other 2 customers, vehicle age was old; hence I found customer's retention ratio was less). He asked a lot of questions on how I found all these things in detail. He asked then: "If a driver's score is added, how do you think that will influence things; what kind of distribution are you expecting in all 3 clusters identified?"

He then literally put my CV aside and started asking questions: He asked about Pandas, PySpark, what exactly is a dataframe, how do you manipulate data, etc. He then gave me 2 problems to perform data manipulation in Pandas, and then I had to perform the same manipulation in SQL (I used MySQL). He was more interested in the exact approach, however. I was given a question regarding students attendance dataset and the key was that he wanted me to use the CASE statement in SELECT statement, group by, having condition and where condition, and there was a non-equi (inequality) join too (2 tables). It was a long query, I had to explain the logic. Then I had to implement the same in Pandas.

By another interviewer: "What exactly is meant by linear regression? What are its basic assumptions? What will happen if each of these assumptions is violated, and how would you rectify that? (I could answer these questions; Chapter 3 of ISLR basically)"

HR questions: "Why do you wanna choose finance and not analytical consulting? Why do you think you are better than the rest?"

Round 2:

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Again, 2-panel interview. Both of them were our alumni. ~30-35 minutes

I introduced myself. They started with economics and repo rate (I openly said that I have no serious background in economics). Then they asked: "What's meant by precision, recall, F-1 score, F-beta score, where is each of these metrics used, what's AUC score, how do you construct AUC curve, how do you find the optimal threshold value using AUC?"

"When can R-square become negative (same as in Mastercard/BCG, same answer)? Why do you wanna choose finance over analytical consulting, what's your motivation therein?"

Given a biased dice, with the probability of a face turning up proportional to the number on that face, find the probability of ... (3 cases were given, don't recall exactly but question was simple; question probably was from Towards Data Science probability page). Then he asked the shooting star problem (Towards Data Science again) (which follows the Poisson process). I was able to answer both these questions.

Question from my time series project: Optimal order selection in SARIMAX. "Why did you wanna forecast the S&P 500 index? What's ADF test, what are the 3 different types of ADF tests? What's the difference between augmented Dickey-Fuller test and Dickey-Fuller test (covered in SAAPM)? Say, you're performing backward feature selection and finally you rejected 10 variables; how will you check that these were the variables not really required (F-statistic is used for this, it's different version; refer to Chapter 3 of ISLR)."

Round 3:

Started with my introduction. A whacky round. They started with my work experience and a case: "Given a car, build a model to identify if it will fail in the next months or not. Which features will you use and why?" (I suggested features related to maintenance, time based vs condition based maintenance, etc)

"Is there any project you would like to tell us about?" (I had talked a lot about my regression and time series project so far, hence I proposed my NLP text classification project) They asked "What did you do, why did you choose LSTM and word embedding approach (I said word size and memory constraints)? Hypothetically, say you have enough memory. Would

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you still go for this approach? (Basic discussion was around bigrams and ngrams which capture the semantic meanings in the words. With them, I could use a count vectorizer, TF-IDF, and other traditional models. However, hypothetically if everything could be fit in memory, there would be issues. Statements like "A eats pizza" and "Pizza eats A" would have the same set of bigrams and ngrams, hampering the classification). What was the model architecture? Why did you use accuracy and F-1 score as performance metrics?"

Then he moved to a puzzle: "5 students are studying under a particular professor who teaches PCM. He conducted one test each for students A, B, C, D & E for each of these subjects. He has marks of all 5 for physics and Chemistry but only 4 marks for Chemistry. Fill in this missing value." (My basic idea was to compare the marks, which were in the 90s, chemistry in 80s and Maths in 70s. Looking at the percentile values, the missing value had 4th rank in Physics and Maths so I took the average of 3rd and 5th guy's marks in Chemistry). Thereafter, we had back and forth discussion on different approaches to solve this puzzle.

18. Rohan Jaiswal [Converted]

(B.Tech in Electronics and Telecommunications from IIST Shibpur, graduated in 2021. Worked for 13 months at EY as an Analyst)

DeShaw **Case Competition:** 1. Time Series Forecasting - Mobility Index of Europe - Time Frame - 2020 to 2022:

2. Finding the right exogenous variables was the key over time series analysis (SARIMA models) - About a week
3. Code format was given - brief comments explaining the logic was important. Had to make presentation
4. Clarity of your approach, technical knowledge & relevance of business insights
5. Highly competitive, it was a difficult dataset, there was a public leaderboard. Shaw focuses more on explainability (so the model should be broadly simpler). I did Random Forest +

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time series (which was something unique). At least 3 slides in the end from a business perspective. (I had cleared it)

6. If you don't qualify here, then resume shortlisting was done

Round 1:

Technical Round

Case Study was about a restaurant in Gurgaon. How would you forecast the revenue (making use of a list of exogenous variables), and which segments will be affected? My approach: First define your scope (what's the time period, revenue channels), then select your exogenous variables (variable categories should be MECE, macro level factors should be pre-thought. In my case, covid related government restrictions/incentives were considered by me; then ask for their thoughts & suggestion on your variables). Every point should be backed by logical reasoning.

Other questions: Why do we have a Harmonic mean for F1 score? Basic Macro questions (Economics: BDM & FRM course should be done before holidays) - GDP, recession, monetary policies. Python queries related to data manipulation - mostly Pandas NumPyp : How do you rate yourself on a scale of 1-10 in python and SQL (rated 8 and 5 respectively): dataset has these columns, and how do you extract a certain output

I would rate this round Moderate to Difficult

Round 2:

Non-Technical Round

Case Study: how would you monetise GPS data for Shaw: start with clarifying questions as to what exactly is the data, which geography it covers, columns, etc (I had Kolkata's cars' GPS data). Need to identify which sectors/areas are being impacted more. More concentration of cars in a region implies more economic activity. Keep asking questions at all levels and it should be conversational. Next, I thought of insurance (a car which is moving a lot is important for insurance companies), and so on. Case discussion lasted for 45 minutes. No analytics

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Other questions: Questions were related to project and mostly on business applications : I had done 3 individual projects in IIM C, one was AirBNB listing price regression project. He asked me to assume a hotel was far from the airport. Would this affect its pricing? (he tried to prove my approach wrong, and he did succeed somewhat.) I agreed that I should have incorporated his concerns in my model. He wanted to check how broad was your thinking, and if nudged in the right direction, did you follow along

I would rate this round Moderate

3rd Round:

HR Round. They asked: "If we roll out the offer, will you accept it?"

My overall experience was Excellent, one of the best interview experiences. Know everything about how Shaw operates. Have good hold on Regression & Time series. I had pre prepared my mindset before the interview. I used to think a lot about exogenous variables and how I would monetise them. PPT skills are very important (need to prepare a decent one quickly)

Rohan's Preparation Strategy: Started preparing while creating the CV. Projects are most important. 50% prep was done with CV prep. Next, identified broad topics (regression, time series, deep, etc.). Studied regression from ISLR (2 & 3) and Montgomery book, time series (SPA link) focused on practical learning via projects wrt time series. ML models from ISLR and statquest videos (thoroughly) and IIM C (Business Data Mining course), Deep Learning IIT KGP (Sudeshna Sarkar slides (topics only) and Mitesh Khapra YT videos, and IIT KGP book is also good). Didn't attend deep learning classes in KGP (prof taught that poorly), Kaggle project is imp.

Inference is most important for statistics. Studied mostly in ISI semester. Imp are CI, Hypo testing, A/B testing, p-interval, power of Test (we could be asked to design our own hypothesis in interviews : null, alt, etc.)

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Case prep: IIM Bangalore case book is very good. Data Science cases are different in the sense that you have to decide on a broad model, which data (attributes) to collect and so on. GDs help a lot here.

Max 4 case comps in CV. Data manipulation is very imp (Pandas)

19. Aravintha Kannan (B.Tech in Mechanical Engineering from NIT Trichy, graduated in 2019. Worked for 18 months as Engineer - Product Development at Mahindra Research Valley)

Round 1:

Technical Round

1. Can Decision Tree handle missing data ? How to handle underfitting in Decision Trees?
My Answer: Yes, which algos can't? Basic linear regression can't handle missing values
2. Explain Yield Curve. : I had mentioned some CFA certification and finance projects in my cv.
3. What happens to bond yield when interest rates go up ?
4. What is Philip's Curve?
5. What are the different types of risk a bond is exposed to?
6. Suggest alternate data for credit risk modelling other than bureau/CIBIL score

I Wasn't able to convey finance knowledge thoroughly. Be careful before mentioning finance related stuff

20. Gaurav Uttam

1. Data visualisation in my work exp?
2. Reg project explain
3. Nlp sentiment analysis project discussion
4. Random forest, bagging,boosting

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5. Regularisation technique

21. Utkarsh Sinha

Case study: Forecast daily mobility index of Europe using statistical / ML / time series models.

Parameters for evaluation:

1. Out of sample model performance
2. Model interpretability
3. Business insights
4. Specific use-cases

Round 2: 10 minute presentation to panel followed by Q&A session.

INTERVIEW ROUND 1

1. You have data of champagne consumption of people across India. What things would you look for in the data to make sure it is accurate.
2. Do you think consumption patterns of any related goods could help in spotting anomalies?
3. What is the problem if there is tracking in regression residuals? How would you solve it?
4. Why do we use harmonic mean for F1 score and not arithmetic / geometric mean?
5. Do you know what F1 beta is?

INTERVIEW ROUND 2:

1. Horse race puzzle. How many races to find 3rd fastest out of 9 horses?
 2. Case: You have data of telephone bills of people in India. How would you monetize it for DE Shaw?
-

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21. Abhirup Saha:

Interview Round 1:

1. Intro
 2. DE mainly focus on time series and linear regression
 3. In TS 1 describe the project - topic was (Forecasting was furniture sales)
-

22. Prakhar Agrawal [Converted]:

Started with a DESIS competition. 6 people will be finalists. All will go to interview on final day

If not selected for DESIS will have to give pre-round

Final day included 2 round and 1 round of Hr. Shortlist people who have done well academic wise. Btech GPA matters. They look for marks and people IR. Not shortlisted an IR whose B.tech was low.

Desis CASE STUDY - FORECAST MOBILITY INDEX OF RETAIL EUROPE MAY 2020-2022, TIME SERIES NEEDED VERY DEEPLY

TIME SERIES & REGRESSION are important

Pre-round (took interview of 25):

1. Explain about work experience.
2. explain work ex in deep (like segmentation, mix strategy)
3. Asked small cases like very small cases.
 1. Consider there is a 5-star hotel in Gurgaon. In Dec 2020(covid) how many people would check in during that period (during dec 2020 covid restriction were relaxed)
 2. Ans - (either corporate, marriage and vacation will happen), vacation are less as restrictions are still relaxing, marriage booking might be the same but audience will be less, corporate events- 0(due to wfh). Give me data from the past months and calculate it.

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3. How would you hedge investment as your investment in tourism,
4. Ans: Bet on ecommerce website as people have money in their, specifically(food items), online food delivery,As people will be on home bet on AC, electricity consumption - electricity distribution,
5. Regression project (not in depth) very less follow question
6. Macroeconomics - what is inflation , recession, etc
7. Given small situation based question like what is impact of Russia ukraine war on inflation
8. What is mean when currency is performing poorly like british pound is performing poorly so whats was reason for that

7 or 8 people selected out of 30

Final Day Interviewer Round1:

Two people 1st one was Swagato sir(Private equity team) & 2nd one was from energy team

1. Case study in depth discussion

Case study: Forecast Footfall in hospital

Clarifying question by Prakhar : which city you are talking about, how many hospitals, give us old data, people come in hospital for which reason,

Ans given by interviewer: covid period 2021,

Ans given by Prakhar: So people might come for vaccinate, people post covid symptom people,covid cases, normal patients

Question asked by interviewer: Tell me about possible exogenous variables

Hint by interviewer: No of people that took insurance in that city so from that your footfall will vary or not?

Ans by Prakhar: Insurance data might be helpful so those who have taken insurance will go more

Hint by interviewer: Govt has given a health scheme so what might be the impact of it. Will that have a positive response or negative?

Interviewers have given many subtle hints and factors and asked whether it might have an impact or not. Total 45 minutes interview. Whole round about structure your thought, it depend on how you catch hints

2. Explain length of stay project (regression project) - Explain impact wise

3. Puzzle - <https://www.geeksforgeeks.org/puzzle-9-find-the-fastest-3-horses/>

D. E. Shaw

Interview Round 2:

Taken by Ankita jha: Purely technical

1. Project (asked about terms) specifically highlighted terms
2. Case: Supposingly you want to open new airport how would you select city what all parameter you would consider

Answer - population, income, frequency of travel, nearest airport

3. Hypothesis testing: You have to test whether citizen of US is spending more than 8k you have to test that

Prakhar's Approach: Started by selecting Null hypothesis, alternate hypothesis, How would he collect the data

How data collection would happen = stratified sampling, In stratified sampling they wanted to know the number of people to take an interview. They wanted to hear it should be greater than 30. They wanted to hear T-statistic, p-value

4. Make graph of income expenditure in USA.

Very basic que of hypothesis testing were asked, very specific technical aspect of terms mentioned in cv were asked.

HR question:

1. What went wrong with DESIS
-

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1. Zara Rahman

Process involved Resume shortlisting and three rounds of interview.

INTERVIEW Round 1:

Following Questions were asked-

- a. Explain kernels in SVM mathematically
 - b. Explain overfitting and underfitting in KNN by taking cases
 - c. Explain the working of XG-Boost
 - d. Explain the gradient step in XG-Boost
 - e. Derive the expression of logit in 2-3 steps
 - f. Give examples of cases where random forest might perform worse than other algorithms
 - g. Few SQL queries and simple python questions were asked
-

2. Rahul N

INTERVIEW Round 1:

Following questions were asked:

- a. How comfortable are you with SQL, Python?
- b. Have you done any projects involving probabilistic models?
- c. What are the loss functions you can use for regression?
- d. If there are outliers they will affect your regression line, what loss function can you use in that case?
- e. How does log transform / box-cox rectify skewness?
- f. Why do the errors have to be normally distributed?
- g. Explain your Photo reconstruction project. How did you use HuggingFace?
- h. What is the MLOps course that you have mentioned?
- i. Do you prefer PyTorch or Tensorflow?
- j. Write a python code which prints all prime numbers till 'n'. Now try making it recursive.

INTERVIEW Round 2:

- a. So what are your current interests in data science?

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- b. If I give you a dataset (x_1, x_2, x_3, y) and additionally give you some causal information about x_1 causing x_2 and x_3 , what will you do now?
 - c. What are principle components? If I give you data with 6 columns and 80000 rows, how do you find it? Why do they have to be perpendicular?
 - d. Does orthogonal mean independent?
 - e. You have two advertising methods, how do you find if one is better than the other? How will you control other factors (one location could be rural that could affect the effects of the advertisements)?
 - f. Suppose you are the data guy of the Indian cricket team. Someone comes and asks you what is the probability that Virat Kohli will score more than 50 in the next match, how will you approach this or what is wrong with this question?
 - g. If I give you all the medical notes (of doctors) with all medical codes of diseases or medicines, how will you build a language model using this?
-

3. Pranjal Sarin

Observation: People with Non-Core (eg-SDE,Analyst,DS,,Pharma industry people) title WEX were majorly shortlisted

Following Questions were asked:

- a. It was more of a random interview because questions regarding DL from Gradient descent to RMSProp, Nesterov momentum, ADAM Saddle points were asked.
- b. Then Question on differentiability of saddle point was asked. I told about hessian, so properties of hessian were asked. If a point is not differentiable what will you do in case of gradient descent.
- c. Then the panellist asked me regarding the Finite difference method.
- d. He asked me concepts -Hypergeometric/poisson distribution, F test, chi square statistic.
- e. Gave me a simple DSA question related to sorting, which I solved.
- f. Then the interviewer asked me how the compiler works in python,
- g. What is Linear Non Linear data structure and how lists are created internally in Python. I gave about 75 percent correct answers but the panellist didn't call me for 2nd rd.
- h. How do you do PCA & How do you do it using SVD

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General Advice:

- Candidate needs to just pray or Know everything (all derivations, DSA, Graph theory, Bayesian Linear regression, etc) and hope that he/she doesn't get a bad panel.
 - (People were asked questions related to GNN with FBprophet/Kruskal /prim algo on paper).
 - Providence panellists don't see your resume properly and decide what to ask so instead of projects concentrate on maths / graph theory / DL and courses.
-

4. Sayanta Midya

The process involved Resume Shortlisting, 2 rounds of technical interview followed by an HR round. Maybe relevant work experience could be one of the criterias for resume shortlisting.

INTERVIEW Round 1:

- Questions were from pandas, sql, NLP and regression.
- What is word embedding? Write the word embeddings of good , better and best.
- Asked to write one sql query using Window Function.
- Is Random forest sensitive to outliers ?explain how feature selection is done using random forest .
- Difference between stack & queue. I was given a data cleaning exercise using pandas and asked to code.

General Advice:

Important Topics - NLP, SQL, Basic Machine Learning techniques (with Bayesian methods)

5. Vaibhav Goyal

Observation was that IRs were shortlisted.

INTERVIEW Round 1:

The first round happened online on Google Meet.

- Highly technical questions were asked,
- SQL questions were asked along with windows Function

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- c. Stat Question: SSD classes will help - SPA is not asked; Inference was asked;

Advice: Keep your technicals very strong. Should be able to apply all technicals studied throughout the year and analyse the interviewer's question properly

INTERVIEW Round 2:

There were Out of the blue questions like:

- a. Consider variables: A, B, C, D, E --- B causes C and D, and D causes E. Take a target variable Z, Predict (Z). How will we do this? This won't happen by Regression.
- b. Correct answer was not expected but focus was on approach:
- c. Told something like Residuals of B&D and B&E...
- d. Direct Concept questions were not asked. Lot of Out of the Blue Questions were asked..

Later, I was called for a discussion with leadership:

- a. Are you comfortable with moving to location
- b. Workplace Conflicts
- c. HR Questions

General Advice:

- a. Focus in Classrooms.
 - b. StatQuest is good for quick Revision - during Interviews but is not completely exhaustive;
 - c. Read Medium - towards Data Science
 - d. Subjects / Electives should be based on interest and not according to the Placement perspective
 - e. Case Competition - Take part in case competition - Helps in PPT creation,
 - f. ISLR is critically important
-

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6. Manvinder Chahar

The process comprised of Resume Shortlist, Two Tech Rounds & One behavioural interview

INTERVIEW ROUND 1 – Done Online ~ 45 Minutes

- a. Covered Breadth of ML & DL| Did Not Go into Depth
- b. Classification Metric-Where to use ? Which is relevant & Where
- c. Classification Errors
- d. Difference b/w Bagging & boosting?
- e. Feature Importance Plot-What are different ways of calculating Feature Importance?
- f. Bayesian Analytics- Taught in IIM is important & asked in Both Round
- g. What do you know about Posterior, Prior & Likelihood? How are they used in Bayes Theorem, Give one application of Bayes Theorem in ML?
- h. Why is Naive-Bayes- naive called naive ?
- i. Pick one Project- Regression Project? What is difference between ridge & lasso Regression?
- j. Which one is used for variable selection among the two?
- k. Tricky Question on Regression Project:
 - i. What is the application of statement: “Three points can not lie on straight line” in Regression Setting: Multicollinearity
- l. Basic Questions on QQ Test:
- m. How do you check for Kurtosis by QQ Plot?
- n. How do you use QQ Test on your project : Regression Project
- o. Live Coding with interpreter:
 - i. Basic Coding Questions:
 - ii. In a string, calculate the number of vowels and consonants in your name? without using the library?
 - iii. count the number of a particular letter in a string without using a function?

INTERVIEW ROUND 2:

- a. What all ML libraries & algorithms have you used?
- b. Graph Questions: We have two events that follow each other. We know the precedence of each other. In a tuple (A goes to B) A must be completed in order to

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start B and so on multiple tuples are present. Now we have to tell the actual order in which events should be completed.

- i. Topological Sort: Used to find this order; Is topological sort based on dfs or bfs - Studied in KGP Sem;
- ii. Same ques was asked to all
- c. System Design : Say every node is a time series in this graph that we have sorted: Design a system to predict the final node's time series value. Write a pseudocode & design.
 - i. Which algo we will use for all these nodes & why not others?
- d. Resume based Ques:
 - i. Graph based project on resume: Describe that project; How does a graph neural network work? What is the role of the message passing layer in GNN?
 - ii. Why do we need another algo for Graph Convolution? Why not CNN?
- e. Asks a lot of Bayseian: Puzzles
- f. Explain the maths behind PCA & SVD.
- g. How will you test the effect of covid in sales of a product? - A/B testing
- h. Why Healthcare?

General Advice:

- a. ISLR & StatQuest
 - b. Focus on courses taught in KGP sem
 - c. Prepare graphs; Bayseian; Mathematics
-

7. Mayank Kale

INTERVIEW Round 1:

Questions were asked related to:

- a. Maths and Statistics.
 - b. CV specific
-

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8. Lagnesh TS [Joined]

The shortlisting for interviews was done based on a resume. The overall interview process was good. There were 3 rounds in total. No tests/case rounds were conducted earlier.

Interview Round 1:

1. What are the components of a SARIMA model specification in time series?
2. Why do we need differencing? How do we decide the order of differencing?
3. What is the difference between ACF and PACF?
4. What is the difference between bagging and boosting?
5. What are bias and variance? Will they be high or low for these models - Random Forest, XGBoost/any type of boosting?
6. In XGBoost, parallel processing is also achieved right. How is it happening?
7. What is the difference between overfitting and underfitting?
8. What is multicollinearity? What is VIF? How is it used to handle multicollinearity?
9. What are the effects of multicollinearity? Tell me about yourself.
2. Why healthcare domain?
3. Why do you want to join Providence?
4. Can you explain about your role in the previous organization?
5. Why do you want to get into data science?
6. What was the difference between your roles in the two organizations?
7. Apart from your current academics, have you used data science/analytics anywhere else in your life be it work experience or your under graduation or somewhere else?
8. Do you have any questions for me?

Case Study:

Consider a patient who got brain strokes is coming to the hospital for admission. At the time of the patient's admission, how will you predict the length of stay (number of days the patient will be in hospital)?

linearity?

10. What is the difference between correlation and multicollinearity?
11. Does correlation and multicollinearity capture the linear relation alone or non-linear relation also?
12. What is R²? What is it explaining? Can it be negative?

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13. How comfortable are you with python? Will you consider yourself to be beginner/below intermediate/intermediate/above intermediate/advanced?
14. Write a python function which will return the dot product of two vectors. Do the dimensionality check also.
15. Write a python function which will return the product of two matrices. You can utilize the already written dot product function if you want. Do the dimensionality check here also.
16. What is TF-IDF? What is the drawback of TF-IDF?
17. What is Word2Vec? Does it capture the context?
18. If a word is occurring in two different sentences, will Word2Vec provide two different embeddings for the same word?
19. What is LSTM? What are the limitations of LSTM?

Interview Round 2:

1. Tell me about yourself.
2. Why healthcare domain?
3. Why do you want to join Providence?
4. Can you explain about your role in the previous organisation?
5. Why do you want to get into data science?
6. What was the difference between your roles in the two organizations?
7. Apart from your current academics, have you used data science/analytics anywhere else in your life be it work experience or your under graduation or somewhere else?
8. Do you have any questions for me?

Case Study:

Consider a patient who got brain strokes coming to the hospital for admission. At the time of the patient's admission, how will you predict the length of stay (number of days the patient will be in hospital)?

Interview Round 3:

1. How were your discussions in the previous rounds? Who took those rounds?
2. Explain about your role in previous companies.
3. How would you allocate the tasks to your team members?

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4. Do you think that those with a computer science educational background have an edge over others?
 5. Why PGDBA?
 6. Why Providence? Why not the finance domain?
 7. Are you comfortable with relocating to Hyderabad?
 8. Do you have any questions?
-

9. Himanshu

Highly Panellist dependent interview

Interview Round 1:

1. Prove OLS by hand
 2. Stochastic gradient descent in hand
 3. Do MLE of normal distribution
 4. Drive cross-entropy loss from binomial equation
 5. A SQL query
 6. Graphs, Crisscross algo, prims algorithm
-

10. Nikhil

Round 1:(45 mins interview)

1. Every aspects we learnt in ISI, IIT
 2. What is logistic regression? Linear regression?
 3. Random forest, probability distribution, random variables
 4. What is inference? Mean, estimation.
 5. ML concepts - bias variance tradeoff, bagging vs boosting
 6. Linear Algebra, SVD, eigenvalues
 7. They asked me everything :(
-

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11. Mukul Mundle (B.Tech + M.Tech in Chemical Engineering from IIT Kanpur, graduated in 2018. Worked for 3 years as Assistant Manager - Digital Transformation at Dr. Reddy's Laboratory. Thereafter, worked for 1 year as Business Strategy Manager - Digital Excellence at Jubilant Group)

Round 1:

Online Round (there were network issues during this round)

They were mostly asking questions randomly

They asked questions on Boruta ("It's a feature selection technique, which is applied above the RF, Gradient Boosting techniques which helps select features").

They asked "What's the problem of multicollinearity and how will you handle it?" (I answered that we could use VIF). I had mentioned Shapley values in my resume. He asked "How would you utilize Shapley values after building any ML model?" (It works on the game theory approach, and it will provide ranking on features with their positive and negative impact on our target variable, and they also give information about the design space. They are used above the ML models).

Then they asked a question on graph neural network (this was driven by their role and its demands). I had done my FADML project on Graph Neural Network (not mentioned in my CV), where we had different compounds (their formula, indicating elements and bonds) and we had to predict whether these molecules were inhibiting the HIV virus or not. We visualized elements as nodes, and bonds connecting them as edges, and approach this as a graph neural network problem. We could utilize a transformer kind of algorithm to solve this problem. They asked how attention would be used for this graph neural network. The discussion went deep into graph neural networks. I gave an answer accordingly, but they weren't particularly satisfied with it. (Last year they had not asked these questions, so this was a complete surprise!)

They also asked some questions on python related to string reversal (I could do that; can't recall the other questions, but they were related to data structures). They had their own platform where we had to code these questions.

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12. Lalit Dhakar [Joined]

3 rounds(Technical, Technical+Managerial, HR)

Interview Round 1:

1. End to end pipeline for a ML project in an organization(because of my work ex)
2. Joint probability distribution
3. A basic python logic involving strings
4. SQL query
5. Deep learning projects(mentioned in CV)
6. Different types of embeddings
7. Sequence to sequence model

Interview Round 2:

1. Why Healthcare(backed it with the work ex I have in healthcare)
2. Care management, The very first project in my CV (most of the discussion was this only around 15 mins)
3. asked me to explain NLP project in the end

Interview Round 3:

1. What led to your parents enrolling you in a boarding school? Was that the right decision?
2. Why Healthcare?

Tips: Try to have some projects from healthcare in your CV and Deep Learning and build story around them

13. R Jayakrishna (B.Tech in Chemical Engineering from NIT Calicut, graduated in 2021. Worked for 10 months as a Software Engineer at Accubits Technologies)

Shortlisting: Resume shortlisting, I had a NLP project on EHR(Electronic health records). Might have been the deal breaker for resume shortlisting.

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Round 1:

What is Regression?

What are the assumptions of regression?

What is correlation?

For a line $y=mx+c$, what is the correlation?

I had a project on NLP, asked what Embedding is, how I go about the project.

Asked me some basic probability questions

Asked me about distributions

Basic understanding of all concepts would be sufficient to clear R1, R2 requires you to think critically, questions varied from graph networks to bayesian inference.

14. Srivatsa Kulkarni (BE in Electronics and Instrumentation Engineering from BITS Pilani, graduated in 2019. Worked for 3 years as a Project Engineer at Wipro. Got to work on basic data related projects, involving things like EDA and outlier detection, for a year)

Round 1:

The interviews conducted by this firm was very random. Many were asked to code using oops in python etc.

1. Asked for intro and then began with SQL query based on joins and group by. Since I had worked in backend development asked whether I know ML model deployment.

2. Asked to implement the gradient descent algorithm from scratch (python implementation part). I explained the gradient descent algorithm and since we cannot use py-torch etc for the case of simple linear regression I told him we can directly compute the derivatives and use them to update the beta0 and beta1. He seemed satisfied with this.

3. Asked to write the formula for Beta estimate using normal equation and then write numpy code for implementing the same (on pen and paper)

4. He asked me to write the logistic loss function and asked me to write the equation of Bernoulli distribution and plot its PMF.

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5. Asked to explain normal distribution and to write down its formula. He also asked something on the lines of “Derive the least squares expression using normal distribution.”

6. I had a project on NLP using BERT, so he asked me what is the max value of sigmoid derivative and asked me to draw the architecture of transformer and explain attention, equations of self-attention & its working, and why use feed forward network. (He was cross questioning me as I was drawing the architecture)

7. I had worked on parallel-processing, asked to write a code to parallelize computing mean, std-dev etc on a large dataset. (question was picked from my work experience; they were insisting on writing the code)

Round 2:

1. He directly began with a graph problem, where we are given traffic congestion data for different time points for different nodes. How will you optimize the traffic flow? I came up with some approach but he wasn't satisfied.

2. He gave another graph problem where using age, weight, height, BMI we need to predict some output and the features have a causal structure. How will you utilize this causal structure for prediction? Since the features are linked the edge weights could be the correlation strength between the two and this would imply that we might have multicollinearity issues. We may have to use PCA to decorrelate the features. He asked to explain PCA and why eigenvectors are orthonormal. He was ok with the approach but more importantly he wanted me to approach it using Bayesian inference (I later asked him about the approach)

3. He saw my DL project and asked to explain it, how fine tuning was done. Then he gave a problem, we are given patient notes and we need to build a model which will be able to answer general questions about the prescription of a patient.

Round 3:

1. Asked the meaning of my name and then a detailed introduction.

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2. He was quite focused on my work experience. He asked about my proud moment in my work-ex, drilled on it.

3. Why did I join Wipro? Why not some other firm?

4. What is the most important thing you gained in your previous organization?

5. Why Providence? Why this role?

Be thorough with derivations, basic formulas of all the PDF, knowledge of DL and Bayesian inference on graphs (must have). Be thorough with advanced SQL.

15. VAM Karthik [Converted]

Round 1:

What is p-value, what is VIF, it's formula etc.

How does logistic regression work?

What is bagging/boosting?

What's Covariance, correlation plot?

What are continuous functions, tests for differentiability, Bayes' theorem and its applications

Coding question: Remove vowels from a string and print the rest

How do you determine if the distribution of data is normal?

Questions about CNN architectures, Relu, leaky relu, Sigmoid, softmax, TF-IDF.

Questions about SHAP, LIME values. What's the difference between them?

BCG

1. Writabrata Bhattacharya [Joined]

There was a case round, had a well structured process | Communication is important in this interview | Good Academic Profile Helped me - Top 20 Percentile

Company organises dinner for shortlisted candidates which is important - Had some questions on Machine learning - Informal Interaction during the dinner

INTERVIEW ROUND 1

Lasted around 20-25 Minutes

Introduce Yourself?

Why PGDBA?

Why CMI & Why Not PhD now?

I was given a case whose details are mentioned below:

Case 1: Walgreen Pharmacy Retailer in US - Facing issues - Doctors in the US provide prescriptions to the nearest medical store. Patients were not going to collect the medicines, & since they had no prescription, they were not taking it from elsewhere. Identify the problem & Provide a solution to Walgreen, so that patients go and collect the medicine.

Interviewee's response to this case:

- Asked questions about how it is spread across various geographies of the US? Is this pattern specific to any geographic location..
- Severity of diseases for which this happened. Age of the patients where this non-adherence was observed. Repeated behaviour from the same patient - Historical Data. Economical Factor - (Not very imp. mentioned by interviewer)
- Gave solution - and asked viewpoints

Advice: Structure the problem - Identify & narrow it down instead of running into solving. Ask as many questions & have multiple perspectives. Keep Your Mind open, take hints. Have good communication skills. Don't be silent for longer.

INTERVIEW ROUND 2

Introduce yourself -

Tip: Having a smile - Communication was critically important.

Case:

BCG

Related to TV Series - On Netflix - TV Series was available on Netflix for some time but in order to earn more profits, TV Series want to shift to Amazon Prime from Netflix.

As a consultant in XYZ, provide a solution to Amazon Prime for Pricing Stranger Things (TV Series)

Tip: Youtube channels for case studies

INTERVIEW Round 3:

Lasted for around 25-30 minutes..

- a. Introduction
- b. Linear Regression & Assumptions of Linear Regression?
- c. Multicollinearity? How do you solve the problem of Multicollinearity?
- d. What is the significance of Beta, p-value?
- e. Explain above things to a layman: Understanding of Basics
- f. What are different loss functions ?

Overall Advice & Tips:

- a. Go Through: IIM-A/B Casebook (imp),
- b. Consider what all features do we have, what data is available
- c. Data Science Case - studies on Youtube (Jay Feng). - Start Now Itself
- d. BCG Gamma / X has channel on Youtube
- e. Test:
 - i. Most Relevant Coding Round:
 - ii. 2 Probability - MCQs
 - iii. Bayes Rule - SPA - Sheldon Ross
 - iv. 50 Probability Questions - Book - Geeks for Geeks Probability Question
 - v. 3 Coding Question
 1. Merge dataframes; Inner Join; Left Join; Pandas; Missing Value Imputation; Changing Datatype of Column; Changing the Name of Column; Pre-processing of Data, Group by
 2. Data Wrangling:
 3. Modelling: Regression / Classification - Use own model - Access to Internet - Use Libraries - Avoid chatGPT & stackOverflow. DataScience handbooks can be looked
 - vi. Practice from Kaggle - Pre-processing & DS Competitions & dataScience

- vii. 6 Machine Learning - MCQs: Courses from IIT & ISI - Reference: ISLR; Dinesh Kumar; Andrew NG's Lectures
-

2. Manvinder Chahar [Joined]

The process started with Pre-placement Talk. The Work is related to working on Analytics Cases or with Consultants There were 4 rounds including a test round. 22 people were shortlisted.

Test round involved:

- a. Aptitude Questions
- b. Probability Questions
- c. 3 Coding Based Question: Jupyter Platform was given to code
 - I. Pre-processing & Data Wrangling:
 - II. Different Dataframe of dataset: Merge dataset to work on one dataset
 - III. Direct Submit could not happen: Too Long Question

Tip: Classification problem was given. Apply one Classification Algo to maximise Recall - I applied XGBoost Classifier. Recall below a threshold was not accepted.

INTERVIEW Round 1:

Following Questions were asked-

- a. Why PGDBA & Not MBA - For Prospective Analysis
- b. Asked about Experience in Tata? Why did you leave your job there?

Was given a Case Study:

A building has a lift, Reduce the waiting time of each individual ?

Discussions happened around below points:

Interviewee: Environment? Users of lift? Is there an alternative Lift?

Interviewee: Are there stairs available?

Interviewer: Only one lift, no stairs,

Interviewee: no. of people on each floor, asked for data? capacity of lift?

Interviewer: Ask for data in a better way? Still not going the crude way?

Interviewee opinion: Maybe interviewer wanted to know Which Derived features can be used in model: Ask data in at most crude form possible.

Lift operation log & Employee Database

Improvise - Lift will wait if there are less people

BCG

Discussion on What is general behaviour of number of push - High in beginning, lower in middle and even higher in end but scattered at end

Questions on Project

- a. Regression Project & TS Project & Classification project?
- b. NLP & DL & Recommender System were not asked..
- c. What's Lasso, Ridge, Scarcity, Diff between OLS & lasso
- d. Random Forest & XGBoost
- e. Flight Price Project : RandomForest & XGBoost
- f. Behaviour of Time Series, Condition of Stationarity & Different Types of Stationarity?
- g. Strong Stationarity & Weak Stationarity? List 4 Conditions?
 - E.g. IID is a strong assumption
- h. What is ADF?
- i. Assumptions for Linear Regression? Correct Structural Form
 - i. Normally Distributed,
 - ii. No Auto-correlation,
 - iii. Multicollinearity
- j. Effect of Autocorrelation violated: p-value will come very high.. How to test: There are two tests

Hypothesis testing:
- k. Classification: SVM, Kernel

Mostly questions from CV were asked

INTERVIEW ROUND 2

- a. What values will you bring from Tata Motors
- b. Are you one of the oldest guys in the batch?
- c. Do you have any issues with the Zero Work Experience guy ?
- d. Telling Politically correct answer
- e. Explain project in TataMotors for Time Series
- f. Case: there is a shopping mart: Order value is decreasing? Identify the issues:
 - i. What does the client do, business model, quantified value of impact occurring, throughout all stores or some specific?
 - ii. Interviewer told that issue is only at one store:

- iii. Interviewee quantified the issue:
 - 1. previous order value & new order value: there was significant drop
 - 2. Assessed Internal Factors & External Factors
 - 3. Discussed issues on Recession / Increase in UnEmployment Rate / New Competition in Market
 - 4. Possible due to Footfall decrease or cart value is decreasing
 - 5. Cart value is decreasing, People are not buying as they used to do.
 - 6. Understand Customer base. Change in customer base ? Bachelor v/s Family Person
 - 7. Availability of certain products in demand?
 - 8. Inventory available, right placement of product?
 - 9. Basic necessity in middle of store, high appealing
 - 10. Psychological Factors for Store to be improved|
- g. Use Time Series Part to predict the Price: Household item are bought in the beginning of month, No sale when sales already high,
- h. To Identify discount labels of each product: on the basis of time-series?
- i. Select some features to predict the discount label of this product?
- j. Market Basket Analysis - Complimentary Products put together, discount on that bundle

INTERVIEW ROUND 3:

- a. Introduce Yourself | Why Consulting?
- b. Case: Suppose a company entering into a stationary space made a pencil whose lengths never change. Help Your Client for Pricing Strategy:
- c. Discussion started on case analysis:
 - i. Asked about Client & its past history / Experience in industry- interviewer told that client is Already in Stationary, but first time in pencil segment
 - ii. Interviewer said it is Normal Purpose Pencil
 - iii. Pricing Strategy: Competitor BenchMarking, Cost based Analysis
 - iv. Value based Pricing: Added to Customer Life
 - v. Gone with Value Based Pricing
 - vi. Particular geography in mind for client: Mumbai as pilot, then pan-India
 - vii. School-going Kids are target user [Kg + 5 years: Student use pencils extensively]

BCG

- viii. Guess Estimated Number of School-goers - Wrong Approach: InCorrect Estimation \ Left in between: New-Approach: How many births are taking place in India - 3 Lakhs were considered..
- ix. How many pencils do they use and amount spent in 7 years? value equivalent to 1750 rupees of old pencils:
 - x. asked for What is the fixed cost & variable cost
 - xi. If the cost of manufacturing is less than 1750, then we can go.
- d. What is Classifier, Random Forest, Logistic Regression?
- e. Explain Regression & its Time Series Application
- f. Credit Card Fraud: List Types of Frauds in Transaction
- g. How are you feeling? How was the interaction last night?
- h. Target Audience for Regression project?
- i. What is Lasso, What is Ridge?
- j. Is it the need of DataSet to use all applications of ML in flight prediction dataset? or you wanted to apply all?
- k. Since datapoint had some inflation point, moved from parametric model to non-parametric model
- l. Explain Metrics of XGBoost
- m. Fraudulent Transaction Detection? Classification Question. Why SMOTE over Random Novel Sampling?
- n. Since it was synthetic data, why trust it? instead of some already present data - Long Discussion on it as SMOTE is not used in Industry -
- o. Use lift or gain stable: Cutoff where lift suddenly falls.
- p. Classification: Boosting & Ensemble Methods: Voting Classifier, Stacking Classifier & its types: Blending.
- q. Soft Voting Classifier & hard Voting Classifier
- r. Why Stacking? Metric Selection Question: Precision vs Recall? Why F1 Score ? To maximise all, stacking was done, instead of F1
- s. Tip: Metrics used to Quantify superiority of any model should be aligned with the business.
- t. F1 score is a harmonic mean of Precision of Recall, WHY HM and not Average? F1 score is closer to lower one

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General Tip:

- a. Hands on Experience is important.
- b. All operations of Pandas, Plotting, Charting, NumPy
- c. Predict & Plot Confusion Matrix without looking syntax
- d. Know basic knitty-gritty of complete process
- e. Data Camp is sufficient for courses: Analytic Case Competition
- f. Kaggle : Any Good DataSet | Notebook
- g. Hands-on ML: CampusX by Nitesh 100Days of ML
- h. Code Imporium for DL
- i. [DL.ai](#) for Andrew NG
- j. Nitesh khapre IIT-M NPTEL for DL
- k. StatQuest - ML Josh Stamer - Best
- l. Cases:
 - i. Booklets by IIM - ABC;
 - ii. Financial News;
 - iii. Flow of Interview - Decides Case
- m. Discuss case among batchmates: form small groups; develop thinking process..
- n. Explain Case to batchmates
- o. Youtube Channel: Case on Point
- p. Puzzles geeks for geeks
- q. IIM - Lucknow - Playlist for cases

3. Pranjal Sarin

Shortlisting was based on Test (Doing prob stats /ml dl MCQs were more imp) I did 1.5 coding questions out of 3 and half of the given theory questions

Test involved many operations questions of Pandas. Do sci-kit learn first.

It was a 90 minute test involving 3 Pandas / Scikit learn questions.

Advice is to first do scikit learn questions because they were easier.

INTERVIEW ROUND 1:

- a. Questions Related to CICD (Bamboo CICD Jenkins etc what will you select) and Deployment were asked.

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- b. Even things that were not in my CV were asked.
- c. What is LSTM? Bidirectional LSTM?
- d. Difference b/w adaboost and XGBoost.
- e. Regression project and Time series project was asked in Detail

INTERVIEW ROUND 2:

- a. I was given an Apparel Recommender system case study.
- b. In which case is content based better than collaborative.
- c. Why is content-based better for Apparel Recommendation?
- d. How will you convert a sparse adjacency matrix into non sparse for recommendation system use?

General Advice:

- a. <https://pbpython.com/groupby-agg.html>
 - b. https://www.tutorialspoint.com/python_pandas/python_pandas_merging_joining.htm#
 - c. <https://www.analyticsvidhya.com/blog/2020/05/datetime-variables-python-pandas/>
 - d. Don't write anything in WorkEx which you have doubts about.
 - e. Keep your resume simple.
 - f. If you have a problem with filling your resume put 2 regression/Classification projects.
 - g. People got shortlists even after doing all theory questions, so do those
-

4. Lagnesh TS

Shortlisting was done based on an online test and resume. I believe the test was for 90 minutes. There were 3 coding questions and approximately 10 probability related questions. The best way to approach is to first attempt the coding questions quickly and then move on to the other questions. Good proficiency in pandas, numpy and sklearn will be helpful for this test.

Interview Round 1:

- 1. Tell me about yourself & walk me through your resume.

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2. Tell me about your work experience briefly.
3. Is your work experience related to data science/analytics?
4. What is silhouette score? How is it calculated? How is it used to evaluate clustering?
5. In a machine learning setup, if pressure and temperature affects a certain variable (output), how will you include both the variables pressure and temperature in the model (pressure and temperature are correlated) to analyze their influence on the output?
6. Do you know LSTM?
7. Have you used it in any project? What is a vanishing gradient problem?
8. Suppose you are using a XGBoostRegressor. On the training set, you are getting a MAPE of 0.06 (6%) and in the test set, you are getting a MAPE of 0.5 (50%). And you can change only one parameter which is maxdepth to improve the test performance. Will you increase it or decrease it?
9. Explain some of the projects which you did in your previous work experience.
10. Compare Software development and Business Analytics.
11. What is multicollinearity?
12. How to address it?

Interview Round 2:

1. Tell me about yourself.
 2. What were you doing in your work experience primarily?
 3. Can you explain the Risk&PnL work which you have mentioned in CV (Simulating source system feed)? Is it a more engineering type of role?
 4. Why consulting?
 5. Why did you want to shift from your previous domain?
-

5. Nikhil Garg:

Test round go for python, basic machine learning

Interview Round 1:

Basic decision tree and random forest and bias variance trade off

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6. Avinash Kumar:

Tips about Round 1 - BCG communication is checked, how answer is presented, give answer by explaining, very imp - communication

Interview Round 1:

1. ML model used in project
 2. XGBoost regression
 3. Project question: Training data MSE - 0.3, Test data MSE - 3.5 what is reason for it, follow up question, hyperparameter used for maximum depth, for improvement of max depth should be increased or decreased
 4. Situation based question(visualisation) -
Company sales data and no of orders given - plot suitable graph for showing it -Used bar graph
Follow up question - what if sales % and number of orders are in lakh
Answer given normalise number of orders, both in same scale
 5. Forecast related question - sales data given, calculated prob in 36 months, whether sales have gone above particular value
-

7. Avinash G

Online test and Coding round: Probability questions, Bayes theorem, ML-related questions, basic data cleaning commands on Python

Interview Round 1: (Online)

1. There are 2 features, which are collinear: how do you use a regression model?
2. Time series - How do you attach probability to the values you predict from time series?

General tips:

Keep your resume simple, don't put complicated stuff

8. Kriti Jain

Interview Round 1:

1. What is VIF and its formula, what does it signify
2. What is PCR, what are the assumptions of PCA

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3. How will you deal if there is interaction present between the predictors
Assumptions of time series
4. What do you mean by variance here, how will you explain it to a layman How will you determine the order for ARIMA
5. How will you deal if there is non-constant variance present
6. What do you mean by seasonality and how will you deal with it

Work Experience related question

Describe the process in customer-integrated life cycle management

Case:

A mid-size office is located in Mumbai, it has only one lift with infinite capacity, what will be the distribution of lift usage? Specify the factors/features that you will consider along with assumptions

Interview Round 2:

1. Explain your work ex-project

Case:

- For a hospitality company, there is a goof up in their pricing algorithm. How will you come up with a new pricing strategy
- Why consulting

Interview Round 3:

- Case: You are a marketing agency and you have a vendor in the Retail space. The vendor produces cloth of different colours, patterns, size, and packaging. The packaging is such that suppose 3 sets of a particular cloth are sold together (say t-shirts). It has been noticed that particular combinations of shirts are sold more and some are not. Now there has been a fire at the vendor side and you as the marketing agency only have a limited amount of stock left - the best-selling packages will be sold out in 1-2 days. Now you have to unbundle the packages so that you have to unpack minimum packages to make the correct combinations
 - What is the distribution of bank accounts in a month
 - An example from real life where normal distribution exists without any assumption
-

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9. Monal Goel [Joined]

(B.Tech in Chemical Engineering from IIT Kanpur, graduated in 2019. Worked for 2 years as Process Engineer [in Operations and Research & Development, majorly on drug development] at Dr. Reddy's Laboratories)

Coding Round: There were 11 questions. 8 MCQs [Probability (Bayes' theorem, expectation), ML concepts (akin to case comps)].

One Regression/Classification problem having 3 parts

1st Coding Question : Data Manipulation (using Pandas), joining tables, finding aggregate values, changing data types, making 1 dataframe from these multiple sources

2nd Coding Question : Data preprocessing _ Data splitting, imputation, etc.

3rd Question : Model Fitting and checking results

We couldn't copy paste from any source. Questions were easy but time was a concern (1.5 hours)

Look up Pandas manipulation; assignments and case competitions would suffice. Don't get into the habit of copying codes. I would rate it 3/5.

Everybody applied and 15-16 were shortlisted based solely on this test

Round 1:

20-30 minutes

It was a technical round

Started with a case study. A company's sales were not doing well and I was asked to find the reasons behind this. I had to start from the business perspective, then I had to ask for data (what all features I would need to solve this problem) and then moved towards forecasting. **My approach** : I started with exploring if it's a demand or supply problem by asking questions. They said it was a supply side issue. Then I moved into exploring as to in which part of the value chain the problem could lie. At one point in the case study, I was told that none of the things that I had pointed out so far was the problem. The company had multiple SKUs. It was a stocking problem. They wanted me to forecast

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why there's a stocking problem. The company's operations were pan India. They wanted me to forecast the stocking. I looked into the features and came up with a SARIMAX model. They asked what's 'X' in SARIMAX model. I said it refers to the exogenous variables and they asked me what exogenous variables I would take.

They asked me a question from my work experience's project (it was analytics related). No business question was asked to me in this round

Round 2:

It was business round (partner round)

Business case study : There was a company into water tanks. There was no growth in this market, and they wanted to move to another industry. I had to find which industry they could enter into. I started by asking questions which would help me understand exactly why they wanted to switch into another industry and what was the current scenario of the market. I looked into what are the current core competencies of the company, and I realised their distribution channel is strong which should be leveraged. Then I analysed some synergies with various other industries and how that can be bundled with the current product. Then I delved deeper into how they could go about building a USP.

Questions on my academic performance : I had mentioned in my CV that I had scored the highest overall marks in ISI semester (you get a scholarship based on that). They asked me which courses were there in ISI, how I did in KGP semester and my current overall rank (although it was mentioned in my CV). They seemed quite happy with the answers

Round 3:

HR round. "What was your previous job like, how hectic was it?" Then they gave me a scenario that BCG usually works for 16 hours a day and if I could manage that, with justification. I gave them the example of my PGDBA experience - despite coming from a non-analytics background, I was holding the 1st rank in PGDBA, which illustrates that I could work hard and manage the technical & other aspects pretty well too.

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BCG interviews weren't pressure interviews as such. They were directing us as and when needed. They wanted to check whether we can answer all questions or not and if not, are we able to pick up their nudges and move in the right direction

Monal's Preparation Strategy: I started preparing 1.5 months before the placements. I started with business case studies (from IIM A book). Then I moved into ML concepts (read ISLR twice). I focused on Deep Learning (as I was interested in AI Garage). I brushed up on ML concepts. I did the bare minimum wrt classes this time.

Materials for case questions: IIM A case book, IIM Lucknow's channels, BCG Gamma online case studies.

Materials for ML concepts : Course slides, ISLR, medium (mostly for Deep Learning), spark by examples website (for Big Data), For doubts : Gooldfellow's book and Youtube channels (statquest, Zed statistics, jbststatistics, ritwik maths). I found UDK rather superficial.

Towards the end, I used to pick one algo from my resume and think in what different ways it could fail and how I could address that

Important subjects : CDS, Inference, SPA (for initial rounds and puzzles), Deep Learning, Big Data (if you have taken the elective and mentioned so in your resume), basic idea of algorithms

General tips : Start with the ML concepts from your CV. For the projects mentioned on your resume, be thorough with all concepts, which other models can be used for similar problem statements, what are the current flaws in your problem and how you can improve it.

10. Anurag Shukla [Joined]

(B.Tech from Indian Institute of Space Science and Technology, graduated in 2019. Worked for 3 years as a Scientist (core electronics) at ISRO)

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Coding Round: 11 questions, 3 based on modelling, 8 MCQs on basic data science and Probability-Statistics (higher level than Case comp questions). Modelling question: Datasets with drivers and vehicle information. 1st we had to collect data from different tables and merge them (joining dataframes based on keys). This was doable but lengthy, so I skipped it. 2nd question was about data preprocessing (missing values, etc.). There was a column on age with missing values, which we had to fill with mean. There were 2 categorical columns that we had to label encode. 3rd question was on model building. Easiest of the three. We had to come up with a model that maximized recall for the dataset. I used a Random Forest Classifier and that worked well.

It was a proper Data Science coding round. We could also use the internet (sans whatsapp & ChatGPT). There were some limits on the time and space the code could take/consume.

Round 1:

~30 minutes. **Business Case Round:** There's a hotel chain in Sydney, and they are struggling to meet their profit targets and their profits are declining since the last few months. They have hired you. Analyze the situation (very open ended).

I started by asking questions: Is this a revenue side or cost side problem, are they paying more (than industry standards) to their employees, are the raw material costs going up, etc on the cost side and what are the different revenue channels for the hotel. Interviewer then mentioned that it's a revenue side problem and asked me to list down different methods a hotel can generate revenue from. Main source: room charges, 2nd channel is restaurant (I was thinking out loud, interviewer was nudging in preferred directions). Interviewer then asked how would you collect data (from whom) for each of these revenue channels. Then he asked how would you quantify these revenue channels. [X number of rooms, avg cost and occupancy of each room; I answered likewise for all channels]. I Whenever I asked a not-so-relevant question (like "Is it a 3-star or a 5-star hotel"), the interviewer counter-questioned me asking why would I need this information, how would it impact my analysis

There was no solution as such to this case. The interviewer was looking for the way we know what data to extract and how we can extract it.

Fit question: "If you take you back to August 2024, when you had joined the course, what would you do differently?" I had mentioned in my interview that I used to read lots of business articles during Covid, which sparked my interest in analytics. I mentioned that I had stopped that after

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joining PGDBA. If I could go back, I would probably continue reading those articles. Then they asked "Is there any course you didn't like and would want to be removed from PGDBA?" I said SSD (in flow) but to justify, I said the topics covered in SSD were some sort of a combination between SPA and Inference. Hence, SSD became quite repetitive.

Round 2:

Technical Round (~30 mins)

CV based round. The interviewer was focused on the words highlighted (in bold) in our CV and he was most interested in the business impact of the projects. My regression project was on flight price prediction, and the first question that he asked was "Why would you want to do Flight Price Prediction?" He further asked "What is boosting, what different boosting methods do you know, why XGBoost is better than other boosting methods, what are the hyperparameters of XGBoost?" (I listed around 4-5 hyperparameters, regularization params, gamma, learning rate, number of trees to be used, depth of the tree, etc; I gave an example comparing Adaboost with XGBoost algorithm). Only cross question herein: I had mentioned that RF is better in adapting to the data than XGBoost; while I had no particular answer to this, I said something related to how the recall parameters of RF adapt to data better (no definite answer to this). I had mentioned OLS, Ridge and Lasso. He asked why 3 different models (I answered: OLS is the base model, Lasso helps with variable selection and Ridge helps prevent overfitting in the data). He further asked "What's Ridge regression, what's Lasso Regression, do they have any drawbacks"

Then he moved to the NLP Project. I had mentioned "Helpfulness of Review" there. He asked what it meant. (Basically, reviews on any ecommerce site cannot be ranked based on the rating therein - a 1 star review could be as important as a 5 stars review. So, reviews should be ranked based on information they provide to customer. Helpfulness is defined thusly - for each product, take the number of reviews each product gets and normalize that with the number of reviews the product has). Further question were on lemmatizer" What's lemmatizer, why do use a lemmatizer (to handle different forms of the same verb, it ensures that root form of the word is maintained), why not stemming (while lemmatizer all forms to the root word, while stemming will convert all forms to a root words which may not be the dictionary root word). I had used TF-IDF as a word vectorizer, so he asked about Word vectorization: what's word word vectorization, which different vectorization techniques do you know (count, Word2Vec, TF-IDF). He asked in detail about TF-

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IDF (which I had used in my project): What's TF-IDF, what's the difference between TF & IDF herein

Some general questions based on key terms in my CV: "what are the different metrics do you know (I answered Precision, Recall, Accuracy, and F1-score), what's the difference between precision and recall (I gave an example of UPI transactions, where we look for precision and bank loan disbursements, where we look for recall), when do you use which one, why is accuracy not always a good measure (not good for imbalanced dataset)."

Round 3:

More of a Formality, Case+Fit round

They started with "Why do you want to join a Consulting firm". In my introduction, I had mentioned that I had been brought up in various cities, and I liked travelling. Accordingly, I answered that I would like to continue travelling and consulting firms would hence be an ideal choice for me, as I would be travelling to prime locations, meeting new people and taking up new jobs. And if I must be in a consulting firm, I'd rather be in a leading firm like BCG.

Then they asked "Why did you leave your previous job?" Again, in my introduction, I mentioned why I joined PGDBA, and the reason was that I had a background in Deep Learning from B.Tech days itself and during Covid, I had read many articles on how Businesses were being transformed using data. I wanted to work in this field, hence I quit my job.

Then we moved to the case: There's Apollo pharmacy in USA, and they have a system wherein a patient goes to doctor who prescribes a medicine over a network which goes to a server, and through that server the customer can choose which pharmacy he/she wants to purchase the medicine from. Usually, patients usually prefer pharmacies close to their homes/offices. Relevant information is passed onto the pharmacy of the patients' choice, and a date is assigned for the customer to visit the pharmacy and pickup the medicine. However, they are facing an issue that the medicine is packed and ready but the customer doesn't show up (Over the Counter money issue - since the medicine is just lying ready, over the counter, to be sold). They asked me to analyze the situation, and how would I promote the patients to show up and buy the medicines. (I was not expecting such a case!) I started asking questions on the processes through which these pharmacies were allotted to the customers., Maybe there's an issue that the allotted pharmacy is too far from the customer. However, interviewers

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answered that this wasn't an issue, as customers could choose the pharmacy. Then I started asking questions that would help me profile the customers who don't show up (maybe young people aren't very serious wrt taking medicines). They asked what factors besides age could affect a patient's decision to buy medicines. I answered if a customer is regularly visiting the doctor, we'll probably have past medical history of that patient, through which we can see for what kind of diseases people usually buy medicines for. Then they asked "Suppose you can do this kind of profiling and predict whether a customer will buy a certain medicine or not. What will you do thereafter?" I answered that if our model predicts that customer is not going to show up, we need not keep those medicines packed." Then they countered: "What if the customer actually shows up contrary to the model's prediction? What will you do?" I answered that this is a supply chain issue, a kind of back-ordering. Companies regularly deal with such issues. Then the interviewer asked "Can you think of some other way in which you can promote the sale of medicines?" I answered: "Apollo Pharmacy has its mobile app. We can send reminders (notification/nudges) to customers to buy the medicines. We can share articles/information about the negative effects of the disease(s) the customer is facing, and how important taking that medicine is." Interviewer was satisfied with this answer and the case ended here.

Then he again asked those questions around "Why did you leave your job?" and the interview ended there.

Anurag's Preparation Strategy: I had attended almost every single lecture, so most of my preparation happened while paying attention in class, particularly in courses like inference & CDS.

I started preparing for the interviews since the last week of the 1-month vacation. For technical revision, the only book that I read was ISLR. I used to watch a lot of YouTube videos, particularly StatQuest. I had an advantage wrt Deep Learning since I was already familiar with them since by B.Tech days. For case studies, I saw the different frameworks used in different types of case studies from the IIM Calcutta Consulting Club's handbook. I didn't solve any particular case as such. Remember that any case should primarily be approached from the business side, and only then should you move to the modelling part (which is anyway the easier part)

I focused a lot on how I wanted to tell a story around each project. So, I had prepared 3-4 lines on the business aspect of each project. When asked to walk through any project of mine, I used to devote a 1st 3-4 lines on just the business impact, sans any technical information. After these 3-4 lines I'd

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dive into the details. For preparing answers for you group project topics, I'd recommend you sit with your groups and discuss the answers.

Revise Inference, SQL, FAD (time and space complexity part for BCG's coding round), Big Data, Deep Learning, Business Data Mining (BDM course itself is a revision). Courses that HRM and Strategic Management could help in HR rounds.

11. Anant Yadav

(B.Tech in Mechanical Engineering from IIT Kharagpur, graduated in 2019. Worked for 37 months at Tata Steel, 12 Months as Mechanical Maintenance Manager and 25 months in Project Management)

Coding Round: There were 5 questions from basic ML & stats concepts; coding test was on data manipulation. Basically, using pandas, numpy & sklearn, we had to perform different kinds of joins, merges, column renaming, aggregations (groupby), apply functions, etc.

Problem statement: 3 different coding questions. Housing datasets in 1st where we had to read 4 csv files, there were technical mistakes in column names which we had to correct based on given description, we had to calculate several metrics; 2nd questions involved datetime manipulation 3rd question was model fitting: we had to fit a very simple regression model (I used RF regressor). Evaluation was based on accuracy, time complexity and how well the code was written.

Total duration was 1 hour. Only 20 students in the entire batch could clear this preliminary round. I would suggest go through Pandas documentation

Round 1:

Technical Round. ~20-25 minutes

Nothing related to CV was asked. Questions related to basic statistics and ML algorithms

"What's meant by PCA? There are 2 variables related to a 3rd variable (A, B with C). You need to regress A&B, which are correlated to each other strongly, on C. You cannot drop any variable. How

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will you do this?" (I approached it using PCR and I how I would complete alpha and beta vectors therein)" He then asked for another approach (standard approach to handle any multicollinearity; it's called error bases correction; basic idea is that all predictor variables are regressed on the variable having high multicollinearity, and then we build a model and compute the errors made by this model). This was a tricky question and took a while. Interviewer asked to describe both approaches in brief ("How will you solve it, how will you get the coefficient estimates, etc")

"What's the difference between PySpark and python? (I had big data as an elective). Define precision, recall and give an intuitive understanding of both these ideas (If a model has a very high precision, then i can rely on that model to correctly identify the positive classes; high recall implies it will make very few mistakes in identifying the positive cases). What is the difference between outliers and leverage points in regression's context? Why is it important to identify these points, and how would you do it? How do outliers and leverage points affect our model (sigma sq estimated will be deviated from the actual if outliers are present)? What's exactly Lasso? Why does it do variable selection? Why does Ridge prevent overfitting? (ISLR answers this very well). R^2 for linear regression is properly defined, but we can still define R^2 if we do not have a linear regression; what's the catch therein? ($TSS=SSR+SSE$ doesn't hold true it's not an OLS linear regression, and hence R^2 may turn out to be negative)

Round 2:

Case study round, given by a partner. ~30 minutes

An optimisation case study: Company "A" was producing product "P" and I had to maximise its total sales in revenue (via total units sold) and also minimize costs. Dataset was available

Approach: You had to form the constraints using regression and optimum function using cost analysis, and use any optimisation technique thereafter. However, I made the mistake of directly starting with the questions, while the interviewer expected me to ask more questions on the dataset. [I started preparing seriously for case studies after this round! There's a very good book: "Case in Point. Complete Interview Case Preparation". It's the source of all casebooks. Use this casebook for preparing the different types of formats and use IIM A's casebook for practice]. Geeks for Geeks has a very good page for puzzles.

Round 3:

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Another partner round. ~30-35 minutes

It started with my introduction. Then he asked "Why do you wanna do consulting; why did you choose this over other options in analytics? Was there any kind of analytics involved in my work experience (I said no right away)? Having done the PGDBA programme, can you think of any improvements you could have done to work work experience? Where could you have applied analytics in your job?"

He then moved to my Assembly Line Scheduler project, where I had build an end-to-end ML model which picked the image of a product and using graph, it identified which manufacturing stage the product is currently in and schedules it accordingly for further processing. Questions: "How did you approach this? How did you get data for this project (I used solidworks to create a trading model and extracted images therein)

Case: There's a pharmacy chain "I" in a country with a rule that doctors must ask the patients regarding their nearest pharmacy while writing the prescription. Each pharmacy has a unique ID number. The prescription is used to share a mail to the pharmacy about the patient's requirements. Pharmacy packs medicines in a box-package. Problem was that despite being assigned so by the prescription, patients weren't turning up to receive the medicine.

Basic approach: what kind of sources do patients have of collecting the medicine. There was no online source and the patient won't acquire it from any other pharmacy as the name suggested by him/her is the most accessible pharmacy for him/her. (This much was clear after I asked a couple of questions). Solution: Doctors prescribing the medication weren't really motivating the patient to take the medicine. Attitude of patients was another issue. Yet another issue was the severity of the disease which would obviously affect their motivation. For minor cases like fever, patients generally didn't bother. They asked me if I built a classification model to identify whether a patient will pick the medicine or not, then the model would give a certain set of predictions. How would I identify the most crucial set of patients to call and which to not call. I said we could sort patients by highest probability and use a Gantt chart to identify a threshold and hence call the number of patients as per my budget.

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12. Aravintha Kannan (B.Tech in Mechanical Engineering from NIT Trichy, graduated in 2019. Worked for 18 months as Engineer - Product Development at Mahindra Research Valley)

Coding Round: Test was for 1.5 hours. Three coding questions and a couple of MCQs on Probability, ML concepts. Attempt the coding questions first, MCQ's are relatively easy, hardly should take 15 minutes. Practice pandas[merging, removing null values, imputation], basic model fitting syntax from sklearn. Attempt and get right two out of three coding questions

Round 1:

Technical Interview

1. Explain how SHAP works : They were expecting more than a surface level answer. They wanted to know in depth how exactly it works. I Had mentioned it in the resume as a project.
 2. Are explainability and interpretability the same ? : extension of SHAP, which is used for explainability. He was confused
-

13. Amrita Mandal

Coding Round: The coding round consisted of case-based questions. There were 4 big coding situation based questions which could be solved by basic knowledge of pandas and basic machine learning packages. The name of the game was speed and clarity. They wanted to know how Swift is with your coding skills. I didn't use much resource, if you had done enough college projects and built simple machine learning regression and classification projects, you're good to go

Round 1:

It was a business case study. My client is a pain India retail company. And they wanted to know why their profit was decreasing. I don't remember it completely but the final problem boiled down to that the client was facing demand forecasting issues and they were over estimating the quantity to be sold and their actual quantity sold was lesser because of which there was over inventory and they were facing losses. Now tackle the problem by suggesting sophisticated demand forecasting techniques like time series analysis.

Remark: Read a lot of case studies and get both technical and business understanding of the cases.

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14. Prakhar Agarwal:

Round 1 - Test

If cleared Dinner with a partner. Check your personality.

Interview Round 1(Purely technical- 25 min):

Until and unless they are convinced they take rounds

1. What is BERT
2. What is transform in detail
3. Explain LSTM in simple language. How are bidirectional LSTM different from LSTM
4. Asked about time series
5. Expected in-depth interview and was kind of stress interview
6. Was cutting in between the answers and not listening whole answer and kind of building pressure
7. Asked about work experience .

Interview Round 2(Case study based):

1. Case study: If an automobile company makes a hatchback car they want to know whether they should enter the SUV market or not.
2. Clarifying question asked: What's their motivation - Expansion. How well is the SUV market doing in india? Two types of SUV - fortuner types and ecosport types SUV.
3. Prakhar created a competitive analysis and explained the case.
4. What you did in last job
5. What impact you made their
6. Why consulting after leaving consulting

Hr round(10 min) :

Basic questions like why you want to join consulting , what were you doing earlier you worked in pharma why you wanted to get out of it. We are extending you an offer

BCG

15. Aishwary Srivastava:

Pre-round 1: Coding test

Interview Round1(Tech round):

1. You got two variables, pressure and temperature and machine performance is dependent on both, both are correlated, how will you remove collinearity & which model you will apply to predict performance of the machine
2. Terms about Deep learning
3. You're in a hospital and your model failed to predict the cancer. Calculate false cancer detection cost. You have to work on precision and recall
4. Five six hyperparameters of XGboost
5. Max depth of Xgboost?
6. In case of overfitting you will increase or decrease the depth of XGBoost

Interview Round 2(Analytics case round):

Case 1: False claim happened in insurance, you have insurance data, you need to reduce the number of false claims without affecting good consumers (those have claim actually)

Case 2: You are a sales manager, the rewards depend on the amount the salesperson does, the reward could be any form, you need to find out which reward contributed how much to the employee. For example if you do 15k sales you would get hawkins coupons, if you do 75k sales, suppose there is another person who already has iphone you tell him that do 75k sale you will get iphone, that is not contributing much, but if you will tell him you will get holiday offer that will get actual incentive, you need to find out/segregate which reward is actually contribution to employee sale.

1. Pranjal Sarin

The process involved 3 rounds and a coding test (DSA based)

Coding Test (Even 3 test cases / 10 clear was enough for shortlist) along with aptitude questions. It was a 1 hour test.

INTERVIEW ROUND 1:

- a. Asked for an introduction and I also pitched about the Regression project done.
- b. Asked to explain the working of Gradient Boosting Regressor/Adaboost in simple terms.
- c. Two panellists asked about the difference between GBDT and XGB, its parameters.
- d. Two prisoners in different cells where they can't communicate with each other, had to toss two coins and guess what the other prisoner got. If at least one of them is correct then they will be released. What strategy do they need to follow ? (Ans: This is not possible)
- e. Bucket Coin Puzzle was asked ?
- f. What is F-test , anova and t-test ?

INTERVIEW ROUND 2:

- a. SQL related Question was asked. It was related to detecting patterns in sequence. (Apply Lag and flag along with window function)
- b. 2 Puzzles were asked.
- c. Culture fit related questions were asked in hindi (Didn't expect this)

INTERVIEW Round 3:

- a. This was mostly a team fit round. It was online and there were 4 panellists, and they introduced themselves first and then asked me to introduce and talk about one of my proud moments. (I told them about DE shaw Desis and AMEX case comp)
- b. Then I was asked about Bitcoin Volatility Forecasting (Time Series) and was asked about exogenous variables that you could have taken.

WWT

- c. Difference b/w GNN based Recommender system (LightGCN) and SVD based recommender system was asked. I explained them in layman terms about how GNN can capture better long term/hop relationship that SVD cannot.
 - d. Then asked about my hobbies and the place where I am from.
-

2. Arpan Singh

The process involved Online test | Two tech round | 1 HR Round

WWT Probability & Aptitude were easier; but Coding Ques. were tougher in WWT as it had Graphs. However they were easier than JPMC.

The test included:

- a. 40 Minutes - for 2 Coding Question
- b. 30 Minutes - Aptitude ~ 20 questions

15 People cleared test

Tip: Leetcode- Easy to Medium; HackerRank - [Pandas & SQL & DSA]

INTERVIEW Round 1: [~35 minutes]

Questions were Based on Resume:

- a. Covered ML Breadth - Regression & its Assumptions & Metrics & loss Function in It?
- b. Normality test Why? If residuals are not normally distributed? Covered from ISLR & KGP Sem
- c. Consulting Company Focuses on Communication & StoryTelling
- d. How can things be explained in laymen terms: Explain XGboost to a layman
- e. Time Series: AR and MA Process? Differences in layman Terms?
- f. Time Series: Why do you need Arch Models. What is stationarity & How do you check for it? Ans: ADF test
- g. Explain ACF & PACF
- h. What is the difference between Bagging & Boosting?
- i. Puzzle: You have 9 balls: 8 of them are equally weighted & one of them is slightly heavy. Minimum attempts to find the anomaly ball?

Tip: Focus on How calm we are? Consulting Firm's culture. Stay Calm & Chill

WWT

INTERVIEW Round 2: [~35 minutes]

Questions were more on coding and puzzles instead of tech.

a. **Coding:** CNN Project: **Image Augmentation** -

- i. How do you resize an image.
- ii. How do you perform image augmentation without using libraries?
- iii. Write code? generally such ques aren't answered.
- iv. What was the flow of project & Libraries used?

b. What is depthwise separable convolution? & its usage in some of the architectures? [Asked because it was mentioned] Not many ques on Gen AI. they work on genAI. if written in Resume. Prepare for it.

c. Clarity on though process of coding questions:

d. Two Puzzels:

- i. There are two Jars: 3 L & 5 L. Unlimited Supply of Water. Measure 4 Litre.
How would you do that?
 - ii. In a test of 100 questions. There is binary marking +1 for correct / 0 for wrong. Two options only for each question. If a person got 75 marks, how many he knew the answer of?
-

3. Ramnath [Joined]

Company looks for cultural fit. Pre-interview test was a qualifying round.

Interview Round 1:

1. 90/120 mins (Coding + Aptitude)
2. Aptitude was typical CAT questions
3. Competitive coding, regular python.

Interview Round 2:

1. Tell me about yourself.
2. Questions from resume & a couple puzzles (Refer Geeks4geeks)

WWT

3. Most part of the interview revolved around resume. Like how did u do it? What techniques did u use. Explain that techniques/algorithms. Explain the business impact of it.
4. Any advice - Prepare resume, hands-on python and puzzles.

Interview Round 3:

1. Third round was a leadership round (Online)
 2. 3 people from US joined.
 3. Asked about myself
 4. What my interest, hobbies are?
 5. Asked me to explain a project
 6. Cracked a couple of jokes
-

4. Emayavaramban A [Joined]

The interview process had four rounds. First online test to shortlist the candidates followed by two technical rounds. Five people made to final round where you have a VC with top management from USA and India

Pre-round:

Pre-round is an online test, where we are tested on ML basics and coding acumen. The shortlisting criteria for second and third rounds seems to be your grasp on basics on ML and AI. Final round selection criteria is whether you can prove that you are culturally fit for the company

Round 1:

The test was for one hour(or 1.5 hrs, not sure). The questions were designed to test the ML and AI basics, coding skills and problem solving skills in general. There is no specific way to prepare for the test. If you are preparing well for placements in general, you will be able to clear this round

Interview Round 1:

For me, first round was taken by Data Science Manager, whom I have seen in the company's Pre Placement Talk. So I called her by her name, instead of calling ma'am. She

WWT

actually appreciated it saying 'Finally someone calls by my name' (they don't expect candidates to be formal just for the sake of it). The interview was about my resume projects and ML basics(when would you use F1 score instead of accuracy, explain some algos mentioned in resume). I was given a problem on probability. I was stuck in the middle of solving it and she helped me out as well.

Interview Round 2:

Second round was taken by a relatively young employee of the organization(whom later I learned is one of the best data scientists they have right now). He wanted to know whether I have proper coding knowledge. I told him 'To be honest, I am not the best coder you will meet today. But I can give you an approach to solve any coding problem'. I was given some problems and I answered them correctly

Problem 1: You have 9 balls and one of them is slightly heavier than other balls. You have a measuring balance to weigh them. What is the minimum steps in which you will find the heavier ball?

Answer:2

Problem 2: A person walks 10 km south, 10 km east and then 10 km north. He ends up in the same place he started. How is this possible? Ans: He is standing in the North Pole. Problem 2 contd: Is that the only possible answer?

Ans: I told that if he is standing at any point in a circle of 10km radius with south pole as centre, it will satisfy this condition as well)

Then I was told to explain BERT in python code (how would you train BERT on your dataset, how would you run the model). I was also asked questions on other resume projects and ML/AI basics

Interview Round 3:

1. Third round is a video call with top management.
2. WWT India's head along with two top managers from USA had a call with me.
3. They introduced themselves one by one and asked me to introduce myself. I was the last candidate to be interviewed so I said 'First of all, Thank you everyone for patiently introducing yourself.

WWT

4. As I understand, I am the last candidate and you guys must be pretty tired of repeating yourself again and again' This immediately put a smile on their faces and my interview was smooth sailing from there.
5. They asked about my previous work ex, why do you want to change your career track, what made you passionate about data analytics, do you know what does a normal day of a Data Science Consultant look like, what do you do in your free time(They also go through your resume on what you have mentioned under 'Interests'.
6. They asked me why did you mention Creative Writing as your interest and I told them I write poetry and short stories and I also narrated a science fiction story I was writing)

General suggestions:

1. WWT expects candidates to be genuine. If you are truly passionate about data analytics and you can prove the same in the interview, you have higher chance of getting hired. They expect people to have a clear understanding of basics and also expect them to have good problem solving skills.
 2. They don't want to hire people who have mentioned fancy terms in resume just for the sake of it
-

4. Diwakar

1. Introduction
2. Bayesian nodes, pretrained RoBERTa model used in project - Give details on how you fine tuned it?
3. What is transfer learning?
4. Simple Puzzle

Tips: As a consulting firm - they were expecting high communicating skills

5. Ayushman Yadav (B.Tech from Mechanical Engineering from IIT Guwahati, graduated in 2020. Worked for 8 months as a Graduate Engineering Trainee at Reliance Jamnagar)

Round 1:

Technical Round based on CV

WWT

Basic questions, don't recall the questions exactly, but they were asking mostly from projects, and they seemed to care not much about the technical knowledge, and focused more on the soft skills

They eventually selected people with work experience and good communication skills.

6. Yogesh Pal (B.Tech in CSE from IET Lucknow, graduated in 2020. Worked for 12 months as an Assistant Systems Engineer at TCS)

There was a screening test, followed by 2 technical and 1 senior management round

Round 1:

All questions were from my CV, my favourite projects, etc.

Round 2:

A panel of 2, both were senior data scientists in WWT. We discussed one of my projects which involved MFCC. Followed by a coin puzzle. Just make sure to communicate well whatever you are thinking while solving the puzzle.

I would suggest preparing puzzles from Geeks for Geeks

6. Srivatsa Kulkarni [Converted]

(BE in Electronics and Instrumentation Engineering from BITS Pilani, graduated in 2019. Worked for 3 years as a Project Engineer at Wipro. Got to work on basic data related projects, involving things like EDA and outlier detection, for a year)

Pre-Round: Basic SDE round test with aptitude and puzzles (I has cleared 2 test cases)

Round 1:

~30 minutes

WWT

1. Asked to introduce myself. Post the intro I pitched my regression project and it turned out to be a good strategy. There was a detailed discussion based on the keywords mentioned in the resume such as multicollinearity, F and T Tests, their null and alternate hypothesis and their test statistic formula. The assumptions of regression.
2. Asked about L1 regularization (picked from my project description)
3. Asked in -detail about the working of Gradient boosting regressor (again, from my resume). How is it different compared to Random Forest? Why did it improve the performance? By doing basic EDA how do you determine if decision tree models are to be used? (If non-linearity is present in the data)
4. Gave an easy puzzle and asked about a project from Work Ex (Analysing Router messages).

Round 2:

~30 minutes

1. Asked for my introduction and I also pitched about the Regression project done.
2. Asked to explain the working of Gradient Boosting Regressor in simple terms.
3. She asked about PCA and K-means clustering algorithms.
4. Two prisoners in different cells where they can't communicate with each other, had to toss two coins and guess what the other prisoner got. If at least one of them is correct, then they will be released. What strategy do they need to follow? (Ans: This is not possible)
5. Since I had a restaurant recommender system project, the interviewer gave a guesstimate to estimate the revenue and factors affecting the profitability of a popular restaurant.

Round 3:

~15 minutes

WWT

1. This was mostly a team fit round. It was online and there were 4 panellists, and they introduced themselves first and then asked me to introduce and talk about one of my proud moments.
2. Then asked about my hobbies and the place where I am from.
3. They asked about a project from my work ex and there were follow-up questions like what the business use case is etc.

Srivatsa's Preparation Strategy: I had prepared a One Note Page for each project, and I used to brainstorm different questions for each of my projects. In the last 10-15 days, I used to revise my CV every day. I focused on DL only in the last 10-15 days. Form a group of 3-4 people and have dedicated case discussion sessions. Develop a good understanding of how businesses run, how they make money, etc. I used ISLR, StatQuest, ChatGPT (Ask ChatGPT questions like "Explain to a 10-year-old how Gradient Boosting works."), Stack Overflow (for questions specific to my profile). Be thorough with probability, hypothesis testing, Bayesian inference and Graphs (last 2 for Providence). Practise doing EDA using SQL

JP Morgan Chase

1. Arpan Singh [Joined]

The process involved Resume Shortlist, Online test, and Interviews: 2 Technical + 1 Behavioral Interview.

The online test comprised of:

- a. Aptitude,
- b. Probability,
- c. Coding Questions: Level was Hard,
 - i. All Objective-C++ &
 - ii. Subjective Test: 2 DSA - LeetCode -Medium level

Overall this round was Medium to Hard:

- a. Similar to Quant Intern Question Test of JPMC
- b. There was a time-crunch:
 - i. 30-40 minutes for 2 DSA ques
 - ii. 30 min for other Ques.

INTERVIEW ROUND 1 (~Lasted for 35 minutes):

Focused Mainly on Resume

- a. Introduce Yourself
- b. Counter Questions on Introduction-
 - i. Hobbies
 - ii. Schooling
 - iii. Family Background
- c. Generic Questions: Choose any one of the Project from Resume: I selected Regression Project & questions were:
 - i. Assumptions of Linear Regression;
 - ii. Asked About Work-Experience
 - iii. RMSE as a metric was chosen. Why?
 - iv. Pros & Cons of Choosing RMSE Metric.
 - v. Why not other Metric

Then, there were questions related to work experience:

JP Morgan Chase

- a. Questions of Work Experience: What coding language was used in Data Scraping for projects in WorkEx?
- b. What library do you know for data-scraping apart from used in projects?
- c. Questions on Python Libraries mentioned in Resume?
- d. How do you handle errors while scraping data from a website?
- e. How can you speed up the scraping bot or optimise?
- f. API related question. How to check if a request from an API is successful or not?
[relevant to work in JPMC]
- g. One Project in SQL: Explain this project & Why did you use SQL & How did it benefit you?
- h. You Have a schema of employees in a company and their salary and their department. Extract the second highest salary person of each department? How would you do this in python using pandas? Pseudo code was asked.
- i. Multithreading for speeding scraping bot. Have you done multithreading?
 - i. Answer was given on basis of previous work-ex: Done using Python

INTERVIEW ROUND 2 (35 Minutes - Online):

- a. Mostly on Aptitude, Puzzles, ML - basics;
- b. Why Analytics?
- c. Regression Setting: If we replicate each observation once. how will the regression coefficients & predicted values change?
 - i. Expected to use formulas for regression coefficient & quantitatively answer how coefficients are changing
- d. When a die is thrown 1000 times, the sum of the number observed is 3400, is the die fair or not? Hypothesis Testing Question
- e. ML - Difference between XGBoost & LightBoost - didn't go into depth, but covered breadth
- f. How to deal with Class imbalance?
- g. Classification Metric: How do you decide which one to choose?
- h. On the basis of Kaggle Project: How did you manage to get this rank & What did you do in that: Stock Forecasting project?
- i. Deep Learning & NLP:
 - i. I had one NLP Certificate: What all courses did you study in that ?
 - ii. Mentioned NER: What is it and How does it work

JP Morgan Chase

- j. What is LSTM & Why do we need LSTM & what is the use of gate in LSTM?

INTERVIEW ROUND 3:

- Questions were on Family Background;
- Learnings in previous Work-ex & How will those be helpful;
- Why JPMC? & Comfortable with Mumbai Location ?
- Gap year: Questions are asked if there is any
- Why Interested in Finance?

General Tips:

- ISLR is very important;
 - StatQuest is important for ML
 - Ritviz Math for Time Series
 - Test is important, relevant work-ex is slightly imp but can work without that too
-

2. Ayushman Yadav [Joined]

(B.Tech from Mechanical Engineering from IIT Guwahati, graduated in 2020. Worked for 8 months as a Graduate Engineering Trainee at Reliance Jamnagar)

Coding Round: Coding + basic aptitude test. General coding questions, 2-D matrix puzzle requiring traversal using coordinate manipulation, etc. They were proper coding questions, not Data Science coding questions

There were 2 main coding questions, 1 was very long and tough (2-D matrix, queen in the corner, you need to save her types), the other not so much (don't recall the 2nd questions)

1 section on general aptitude (probability, etc.) and 1 section of Maths aptitude (basic class 9-10 level geometry, algebra questions).

Total time was 40-45 minutes. I could solve 6-7 in Maths/10, 10/20 in general aptitude and half of the 2nd question in the coding round (it passed some cases, not all of them).

JP Morgan Chase

11 students were shortlisted via this coding round

Interview Round 1:

Technical Round

They went through the CV, traversed through the key words. They asked if I had done any project on time series. I had done 2 projects, using LSTM in one. They asked: "What LSTM, how does LSTM work, what's the advantage of SVM over Logistic Regression (support vectors are robust to outliers and they depend only on the points inside the margin and immune to outside point, but we do need good amount of data for proper training), what's TF-IDF, what's the use of TF-IDF, what information does it convey?" They were asking basic questions, and didn't dive deep into the concepts, particularly in Deep Learning.

Few python questions: "If you copy a dataframe, is the whole dataframe copied in python (i didn't know about python, but knew that there are copy by value and copy by reference values therein. I made an answer accordingly). What's deep copying and shallow copying in python (I didn't know this)

Probability question: "You have some coins, 20 of them are heads, what can you do to ensure that half turn heads and half of them turn tails." It was a Geeks for Geeks puzzle (trick was something like flip 50% of the coins). I couldn't answer it. Then they asked: "There are 3 washrooms, one is occupied 30% of the time, other is 20% of the time. If you could use a washroom, what's the probability that it's the 1st one?" (Bayes' rule question).

No business question was asked

Interview Round 2:

HR Round

What do you expect from this role (I referred to their pre-placement talks and projects discussed there to answer this question), why JPMC, why are you fit for this role (pre-prepared answer, 1-2 lines about JPMC being the oldest brand, interest in working BFSI sector), what other interviews did you sit for (I said DE Shaw, since I was interested in finance)?

JP Morgan Chase

My interview was decent and I converted, they were listening to my answers and didn't ask too many questions in between

Ayushman's Preparation Strategy: I started in July break. ISLR for classification, regression, SVM, etc. Then I prepared some of my projects thoroughly (anticipating their questions)

I tried to read on the details of the concepts used in my projects in the last 10-15 days

For case studies, IIM A, FMS, etc. case study books. I used the internet (including ChatGPT) for Deep Learning. I revised the 1st 4-5 chapters of Sheldon Ross for probability.

Prepare DL topics well. PPTs are very important in placement case competitions, particularly for consulting roles. DO NOT bluff in your CVs randomly

Oliver Wyman

Sayanta Midya

The process involved resume shortlisting, a Case Study on time series forecasting followed by 2 interview rounds.

The Case Study was on Time Series Forecasting of Covid Death prediction of 4 countries. I personally devoted 3-4 days to it. Be thorough with all the machine learning techniques and make sure your PPT is good.

INTERVIEW ROUND 1

- a. Difference between precision and recall
- b. Gave two puzzles
 - i. One on bayes theorem
 - ii. Another famous tiger and goat puzzle.
- c. Basic questions on regression and time series.

General Advice:

- a. Is there any other thing you want to share about the process?
 Ans. You should be very crisp and clear when you answer, be thorough with the fundamental machine learning algorithms and finally you should be confident when you answer.
 - b. Geeks for geeks puzzles,
 - c. Case study practice from OW website
 - d. Victor Cheng for Case Studies
-

1. Priyanshu

There were a total 3 rounds, one involving a case study on Time Series Data. It was related to No. Of covid Deaths with time for 4 different countries

Forecast For Next Months -

Time given for case study: 1 Week to Analyse

Focus: Presentation Skills & Coding Skills - Clean & logic

Around 25 people were shortlisted on the basis of this case study.

INTERVIEW ROUND 1:

Started with Introduction followed by Little Discussion about Work-EX in Axis Bank & India Today

Oliver Wyman

Following questions were asked related to Academic Projects:

- a. Regression Project - Mean absolute Percentage Error: 0.22 which isn't great -So Interviewer Asked Ques Regarding That -Why is it Still In Resume? Asked Probable Reasons why is it coming?
- b. Interviewee said that a limited Number of Features were available, Gave Honest Answers; Other Models were also tried and 0.22 was best obtained.
- c. Discussion on Myers Brigg Algo - Classification Project To Optimise Classroom-Engagement.. - Novel Idea - Not implemented - Was Pivotal
 - i. Explained In Depth - Thought Process behind it - Shared idea for around 8-10 minutes.
 - ii. Cross Questions on Project: Thought Process - Increase Attention Span of class - Case where this Model will fail
- d. Light GBM & XGBoost - Explained what they are & Difference between the two? - Interviewer (a PGP alum of IIMC) asked to explain in Layman language with an example?
Tip: Light GBM - Include features like exclusive feature bundling & Gradient One Sided Sampling in your example
- e. NLP Project was discussed: Sentiment Analysis
 - i. Pharmaceutical reviews were analysed and sentiments were predicted.
 - ii. How would you convince me to use this model?
 - iii. Complete discussion around the business?
 - iv. How to convince doctors to use this model?
 - v. Cross Questions- Showed use cases for doctors where the model will be effective for them.

INTERVIEW ROUND 2:

The interview was taken by HR & Head of Data team

- a. Gave Introduction - Non-Conventional Approach: Others were asked to mention things not in resume;
- b. Discussion on Myers Brigg Project: 15 Minutes

Oliver Wyman

- I. Covered Everything From Thought Process to How to Tackle Various Cases?
- II. Included Technical & business ques

Discussion was initiated on a case study whose detail is as below:

Problem Statement: Store Chain wants to open a new store in location Y. Research shows that Morning buyer buy twice than evening buyers; female candidates buy 1.5 times than male candidates;

I. There is a salesman who will distribute pamphlets? How will you position & Whom will you target?

General Advice:

- a. Don't put ML into everything, Data Science case makes no sense without Business
 - b. Instead Ask Interviewer: If they want us to explore ML angle to that business case..?
-

2. Robin Singh

The process had three rounds including one case round. Case was given a few days back. - Mid Aug. (Around 14 days back)

Case involved Covid 19 Death Data - Predict Daily Count of Covid-19 Data for next month in 4 Countries

Tip: Consider & Explore Exogenous Models: External Data - Temperature, Humidity, Hospital Beds, etc.

It also had Missing Data for some countries - Data Cleaning & Wrangling & imputation was required.

INTERVIEW ROUND 1:

- a. Tell me About Yourself;
- b. How did you do population prediction? - Work Ex Related Que. & Resume based (interviewer was also civil engineer).
- c. What more features could be used to predict the population of a region?
- d. If any project is mentioned in the resume or discussed, business problems, features in the project, what could more be thought of, rectified, and models used should be well prepared.

Oliver Wyman

INTERVIEW ROUND 2:

Following questions were asked:

- a. Classification Project on Fraud Detection:
 - i. Two methods were compared for accuracy and precision
 - ii. Why Precision and Not Recall? - Compare both merits - and all scenarios
- b. Regression Project: (Favorite Project) - Airline Ticket Price Prediction
 - i. Which other features could have been considered.
 - ii. Why not consider Dynamic Pricing?
- c. Audion Classification Project:
 - i. Mentioned Accuracy, Precision, Recall- Why mentioned all three? All three were not required.
 - ii. Data was balanced
 - iii. How was the F1 score calculated? Is it for all instruments or one instrument?
- d. Deep Learning - OW can go into it. Asked questions about CNN? - Definition only

3. Avinash Kumar:

2 round 1. technical 2. kind of hr and tech mix
case study shortlist - individually

Case study: forecasting of covid 19 death in 4 countries(You are provided with historical COVID-19 death data from January' 21 to May' 21 in the file Dataset.xlsx. The goal of this case study is to predict the daily count of COVID-19 deaths for the next month of June' 2021 in Bangladesh, Bolivia, Brazil, and Bulgaria. We encourage you to make use of external datasets, explore various methodologies, and generate valuable insights.)

Interview round 1:

Intro

Discussion on Classification (credit default prediction)

Clustering (bank customer segmentation)

Recommendation system (embeddings, graphs)

Oliver Wyman

How did you answer - question was on project, project question was based on problem statement, features, whatever wrote on cv like decision tree, , general flow of project from start to end

Bias variance tradeoff,
 multilayer perceptron
 formula for weight update
 loss function eqn
 bayes theorem numerical question

Suggestion : Focus on tech, business problem case study, look from different angle , ask relevant clarifying questions , ask time from the interviewer.engage them , case study discussion happens in both direction you and interviewer.

4. Diwakar

15 shortlist for interviews, 2 panels (One panel is going like a stress interview)

Interview round 1:

1. What is the difference between correlation and covariance?
 2. What is i.i.d, its assumption?
 3. Case: there is a banking firm, when a user complains on a fraud transaction if the fraud amount is less than \$5, they're simply paying the amount to the customer, for the greater amounts they escalate the issue and taking actions. They noticed that the fraud complaints cases with amounts less than \$5 are increasing and the bank is paying much. What will be your work around this to solve the matter?
 4. (He asked if he can get the data of customers reporting these complaints, they said the firm won't disclose it. They were expecting a business solution not an analytical one)
 5. For some ppl, they asked to explain the case given in shortlisting round
-

Oliver Wyman

6. Mukul Mundle (B.Tech + M.Tech in Chemical Engineering from IIT Kanpur, graduated in 2018. Worked for 3 years as Assistant Manager - Digital Transformation at Dr. Reddy's Laboratory. Thereafter, worked for 1 year as Business Strategy Manager - Digital Excellence at Jubilant Group)

Pre-Interview Case Study: They had given a case study prior to the interview. We had 3 days' time. It was an individual case study. I was given Covid deaths dataset for four for 5-6 months countries based on which, I had to predict covid deaths for the next month. So a time series problem

I used exogenous variables like covid cases, hospitalisation, stringency index, level of control by different governments for this case study

Submission required a ppt and well made code notebook

Round 1:

They asked "What's bias-variance tradeoff, what's overfitting and underfitting?" (general question, not picked from my resume)

They gave a case study where we had a median company (like Star Sports or Sony) as our client and they had to bid for a cricket tournament scheduled in 2028 (5 yrs down the line). We had to suggest a bid price. So it was a profitability kind of case study. My approach was to find the break-even price first. First of all, we had to identify the sources of revenue from the tournament - share of ticket prices, advertisements, merchandising, etc. When I mentioned these revenue sources, I was given revenue values for each of these sources. Then I moved to expenses - infrastructure (IT). He shared the corresponding value. Based on these, we can calculate profit after 2028, which we had to discount to 2023 (time value of money, he gave the interest rate too). That would be our bid price. Interviewer seemed pretty much convinced with my answer

He had also asked a simple python question, and asked me to solve it without pandas (I can;t exactly recall it)

Round 2:

They started with my introduction, and then asked about one of my work experience projects: It was a data science project on pharmaceutical industry, where we wanted a good yield of the products.

Oliver Wyman

In the manufacturing setup we'd have different unit operations' initially we'd use certain raw materials - active pharmaceutical ingredients, supporting ingredients - and finally we'd get the tablet. Yield is output/input, and we had to maximize yield. Since these raw materials are going through different unit operations, there are different parameters included in this product supply chain, parameters like material attributes, process attributes, human attributes (which person is operating which machine, temperature and pressure of the machines, etc.), quality parameters. This was basically a regression problem. I explained the data sources, different variables involved in this problem statement and how I approached this problem statement. They asked questions on performance metrics and outcomes of this project.

Then they moved to a case study: There's a case study about an existing sports store in a sports arena, and another sports store had to be opened in another part of the city. Interviewers had given me information about the customers of existing stores, when they were visiting the store, % of women and student customers. We want to open a store near a different sports arena, and market the same using pamphlets. Which section of the population should these pamphlets be distributed to? My approach: I started with questions on different types of customers visiting the existing store. I provided a solution from the RFM point of view. Idea is that we can bucket overall customers into buckets of users who are visiting recently, frequently, their monetary value and we can cluster users of existing store based on RFM technique, and we can apply learnings therefrom into the new store. Besides, we can do some kind of Porter's Five Force Analysis for the new store which we are opening - no. of competitors in that particular area, behaviour of customers visiting that area, etc. Post doing all these analyses and clustering (using RFM), we can distribute pamphlets to the most promising set of customers.

I had mentioned in my CV that I was into adventure sports, so they asked a generic question related to that.

[When I was done with this round, I was told that I had prepared very well for this round. However, I feel I was judged on the case study asked here and hence not selected]

Oliver Wyman

7. Aravintha Kannan [Joined]

(B.Tech in Mechanical Engineering from NIT Trichy, graduated in 2019. Worked for 18 months as Engineer - Product Development at Mahindra Research Valley)

Pre-interview Case Study: Time-Series Forecasting on COVID-19 Deaths. [PPT+ Code Submission]

Time given: 5 days . Individually done, used ARMA & XGBoost

Focus on Data Pre-processing, features for modelling and the slides (PPT) should be more visual than text with business implications than just model.

Round 1:

Technical Interview- Focused only on projects and in-depth understanding of highlighted terms in resume.

Few Questions:

What is the significance of doing PCA in K-means clustering? My Answer: PCA is used for dimensionality reduction

What are the consequences for explainability? My Answer: PCA takes a linear combo of features and gives a new feature, but it's not easy to interpret. So that's the problem. New features should be interpretable

How is XG-Boost different from normal gradient boosting? : Standard answer

How do you decide when to ridge or lasso regression? My Answer: Lasso for feature selection & ridge for too much multicollinearity. OW didn't go deep

Given a chance again, what additional features would you use for modelling this problem? (applicable to any project) : asked wrt a project. Told them additional features

Round 2:

Oliver Wyman

Case Study Round: A business case on customer targeting strategy for a sports company was provided and the candidate is judged based on understanding of the business-problem, breaking it down and approach of moving towards a solution : It was more of a business case study -> kids come to buy sports equipment at a sports shop near a playground. How do you know which customers to give discount coupons to? -> Ask about the profile of the customers, who's buying more, seasonality, just get more information. Didn't get into any analytics

Asked about my Favourite project : They wanted to check how excited I was wrt my projects and was there any genuinity. I explained it very well (it was from EXL case comp)

Questions about the Case competition : What features did you use? Why do you think you made it to the final round? Why didn't you win ? My Answer: Be honest, others were better at presenting, etc.

Suggestion: Ask clarifying questions. Do not jump right into the ML Modelling aspect directly.

Understanding of business aspects of data science, Machine Learning concepts and Statistical Methods are important . Effective communication is crucial, so focus on that as well.

Aravintha's Preparation Strategy: I prepared for my interviews during our academic break in the third semester at IIM-Calcutta.

Recommended Materials:

1. ISLR book and Paul Martin's summary notes
2. Business Analytics :The Science of Data-Driven Decision Making by U Dinesh Kumar for chapters on Hypothesis Testing, Logistic Regression.
3. Jay Alammar for BERT, Transformers topics
4. Medium blogs for highlighted key terms of your resume projects.

Important Topics: Regression, Decision Trees, Boosting, NLP, Clustering, Statistics & Probability

General Tip:

Write a simple resume with projects you feel confident about talking about in an interview, rather than throwing complex technical jargons.

Oliver Wyman

8. **R. Jayakrishna** (B.Tech in Chemical Engineering from NIT Calicut, graduated in 2021. Worked for 10 months as a Software Engineer at Accubits Technologies)

Round 1:

Difference between bagging and boosting, What is correlation, name a structure which can be correlated and have 0 value for correlation(circle). Puzzle, case study.

9. **Srivatsa Kulkarni** (BE in Electronics and Instrumentation Engineering from BITS Pilani, graduated in 2019. Worked for 3 years as a Project Engineer at Wipro. Got to work on basic data related projects, involving things like EDA and outlier detection, for a year)

Pre-Interview Case Study: Time series based Covid deaths prediction for 4 countries. Approach: Get the variables that will help predict deaths (I created few lag variables & exogenous variables - such as occupied ICU beds; I also used ChatGPT to brainstorm these features). Then I did basic EDA and plots, then my baseline model was linear regression and then XGBoost. I also used Shapley plots to interpret the features. However, my business recommendations were bad (I didn't properly think them through)

Round 1:

1. Introduce yourself and then they picked up one of the case competitions and asked about the business problem and features involved. They asked about feature engineering we had done and to develop at least one new feature using existing ones. There were no questions regarding ML models used.

2. They asked me to explain the classification project, features used, what is IQR?

3. Asked to explain my recommender system project. They questioned a lot on how does Alternating Least Squares based matrix factorization solve the cold start problem. I felt the panel was not very much convinced with my answers but they kept on questioning me.

Oliver Wyman

4. Customer segmentation project. Asked to explain PCA and what are the specific insights generated using it. They were expecting in-depth analysis using PCA and more on business insights. The panel was very critically assessing the answers given and there were lots of counter questions

Be very thorough with the projects and case comp written in your resume. For me the panel asked questions from projects aligned more towards business such as segmentation and Recommender System.

Renew Energy

1. Vaibhav Goyal [Joined]

The process involved two interview rounds.

INTERVIEW ROUND 1:

First round lasted for 15 minutes and following questions were asked:

- a. Puzzles were asked - geeks for geeks were asked
- b. Technical questions:
 - i. Random Forest;
 - ii. XGBoost;
 - iii. Linear Regression;
 - iv. Things taught in IIT / ISI were asked.. Out of the blue will not be asked..
- a. Previous Workex was not asked much.
- b. StatQuest will be sufficient to answer them;

INTERVIEW ROUND 2:

- a. Previous Work Ex was asked.. If has worked previously with data or not
- b. What is XGBoost, Random Forest.
- c. What is the difference between the two; Bias, Variance & tradeoff was discussed.
- d. Which case will XGBoost perform better and same for Random Forest?

General Advice:

- a. Not much out of the blue;
 - b. Focus on classroom studies will be sufficient to crack;
 - c. Handle confidently
-

2. Lagnesh T

Shortlisting was done based on CV.

Interview Round 1:

This was a technical round. This round didn't involve any case studies.

Questions asked:

1. Tell me about yourself.
2. Why data science?

Renew Energy

3. Are you comfortable in python? Rate yourself out of 10.
4. Asked to write code for the following: i. Create a list containing only the second last element from another list. ii. Create a list containing the last two elements from another list. iii. Create a list containing the second element and second last element from another list. iv. Create a list containing the second element and last two elements from another list.
5. Write a program to identify if a number is prime or not. How do you optimize this further (number of iterations)?
6. Three ants are sitting on three corners of a triangle. They all start at the same time with same speed. They can go in any direction in the triangle. What is the probability that they will collide?
7. What is SARIMA? What are the additional components to represent SARIMA apart from ARIMA?
8. What is the difference between ACF and PACF?
9. Look at this ACF plot, what do you infer from this? Is there any correlation between the lags?
10. What is the difference between bagging and boosting? In case of high bias in our model, do we go for bagging or boosting?
11. What is LSTM? How do you tune its hyperparameters? What are all the hyperparameters? How can you avoid overfitting in LSTM?
12. What is the difference between overfitting and underfitting?

This round was moderate.

Interview Round 2:

This was a technical round. This round didn't involve any case studies.

Questions asked:

1. Tell me about yourself.
2. Pick any one of your projects. Explain the problem statement, methodology, approach and the results. Why did you go for a linear model in the project?
3. Puzzle: There is a number generating machine. It can perform only three operations: $x+5$, $x+7$, $\sqrt{x}$. Currently the number displayed in the machine is 0. Try using these operations (any number of times) for the machine to generate 2, 10, 14. The following conditions need to be satisfied during the execution. Otherwise, the machine will stop working:

Renew Energy

- a. Any number should not be generated repeatedly (more than once).
- b. The generated number should not go beyond 60.
- c. The generated number should be an integer.

Come up with an approach (process) so that it can be programmed. Exact solution is not required for the specific numbers (2, 10, 14). Tell a generic solution.

- 4. Are you okay to work in Gurgaon? Or do you have any location constraints?
- 5. Do you have any questions for me?

This round was moderate.

Interview Round 3:

This was neither a technical round nor a business round. This round didn't involve any case study.

Questions asked:

- 1. How are you? Where are you from? Tell something about your city.
- 2. What do you know about our company? Why do you want to join?
- 3. Are you okay with working in Gurgaon? Or do you have any location constraints?
- 4. Do you have any offer with you now? If we give an offer, will you be joining us?
- 4. Where do you see yourself in the next 5 to 10 years?

This round was easy.

3. Avinash Kumar:

Interview Round 1 -

- a. Intro,
 - b. One puzzle - <https://www.geeksforgeeks.org/puzzle-27-hourglasses-puzzle/>
 - c. Things not written in cv ,
 - d. explain NLP project,
 - e. Python Coding - Create list of 5 items , print 2nd last element
-

Renew Energy

4. Gaurav Uttam

1. Intro
 2. Case comp basic discussion
 3. Logistic reg work exp project discussion
 4. Assumptions of logistic reg
 5. Precision, recall, f1 score, roc curve
 6. In medical which performance metric should use (situation given)
 7. Write Python code on reversal of list
-

5. Utkarsh Sinha

Resume based shortlisting

INTERVIEW ROUND 1:

1. Puzzle - Monkey throws a dice. If the number on the dice is 1 -5 , he eats no of bananas corresponding to no on the dice. If it is 6, he eats 5 bananas and throws the dice again.
2. Expected no of bananas the monkey will eat?

INTERVIEW ROUND 2:

1. Assumptions of linear regression?
 2. 2 residual plots
 3. First One against no. of obs and second one against time
 4. First one - straight line above x axis with width const
 5. Second one - parabolic shape with width const
 6. What are the problems with residuals in this and how to fix them?
 7. Variance seemed constant as width of residual plot was constant.
 8. Plot against time suggested tracking in residuals. Since it is parabolic, may need to do second differencing.
 9. Had bias (mean not equal to 0)
 10. Mathematical definition of sample and population?
-

Renew Energy

6. Srivatsa Kulkarni (BE in Electronics and Instrumentation Engineering from BITS Pilani, graduated in 2019. Worked for 3 years as a Project Engineer at Wipro. Got to work on basic data related projects, involving things like EDA and outlier detection, for a year)

Round 1:

~20 minutes

1. Asked to introduce myself. I pitched my regression project as well (I wanted to drive the interview towards my regression project, since it's easier. Hence, I used to mention my regression project in my introduction)

2. Based on the project he asked various questions about F and T test. Asked about their null and alternate hypothesis.

3. Probability of an accident in any given hour is $1/2$. What is the probability of an accident in any 30 minute window?

4. He asked a simple probability puzzle like

- a. Probability of having both children as girls given that the first child is a girl.
- b. Probability of having both boys when at least one of the children is a boy.

(I was able to solve these probability questions, although it took me some time)

5. Try to engage the interviewer when solving the problem. Tell them your approach and just take this whole process as a general interaction and be yourself.

Round 2:

~20-30 minutes

1. Asked to introduce and explain one of my work experience project

2. He asked me to write a python program on paper to determine whether two given numbers are coprime.

Renew Energy

3. If a bus contains 100 seats and 100 tickets are given 1,2,3...100 corresponding to seat number 1,2,3...100. Now the first passenger with ticket 1 has lost his ticket. He takes a seat randomly and not in seat 1. Find the probability that the 100th passenger will take his seat (100th seat).

Round 3:

1. It was a general interaction round. Asked me what is special about my city. Asked about my hobbies.

2. Whether I will be ok with relocating to Gurgaon.

3. Asked why Renew Power? And what do I know about the company?

Know more about the company, be good with basic hand's on in python and also practise moderate probability and combinatorics problems. Go to the interview and have a natural conversation where you genuinely take interest in what is being asked, take a moment before answering and it's ok if we make a mistake as long as you are trying to solve the problem without losing balance.

UBER

1. R. Jayakrishna [Converted]

(B.Tech in Chemical Engineering from NIT Calicut, graduated in 2021. Worked for 10 months as a Software Engineer at Accubits Technologies)

Round 1:

Interviewer: Introduces himself, asks me for an introduction

Me: Introduction

Interviewer: How much would you rate yourself in Python, Pyspark and SQL

Me: No exposure to PySpark, python 7.5, sql 7.5

Interviewer: Proceeds to case study, tells me what uber's business model is. Then gives me a case.

Case- All app-based cab hailing services in India are facing a new challenge, Drivers are calling up customers after a ride has been confirmed, they request the rider to cancel the booking and asks them to pay upfront. Such trips are called offline trips. Present two hypothesis why this is happening.

Me: Reiterating the question, asked a few clarifying questions. Later on, presented two hypotheses.

Interviewer: Interviewer acknowledged that both the hypotheses were good. He asked me how we can know if an offline booking has been made.

Me: Told him how the data can be used, some feature engineering

Interviewer: was giving feedback and counter arguments for the methods I mentioned

It was past 30 minutes

Interviewer: Let's wrap up the interview. Do you have any questions for me

Me: Asked a question on the models used in production and if Bayesian based models are used

Interviewer: told me he isn't aware of it but told me that there's a good chance it can be used

UBER

Round 2:

Interviewer: Introduces himself, tells me that another person will join the call in the middle, asks me to introduce myself

Me: Introduction, also tell him that regression and time series are my favourite subjects

Interviewer: Asks me to explain my regression project. Asks questions on F-test, Multicollinearity, assumptions of regression. Finally asks questions on R^2 . What is a good value of R^2 . How do you explain this to a Businessperson? What are alternative metrics? What is RSE, what is MAPE. Advantages of these metrics.

Interviewer: case

Case- Cab hailing apps charge a cancellation fee under certain conditions, what are these conditions? When can a cancellation fee be charged on customers and on drivers?

Me: told him certain conditions when cancellation fee can be levied.

Interviewer: Now that we've decided under what conditions we can charge a cancellation fee, how much should the cancellation fee

Me: Told him how different services can be charged using different multipliers.

Interviewer: How would you know which policy is working better if you have 2 policies.

Me: Told him about A/B testing.

Duration- ~40-45 minutes

Round 3:

Interviewer: How were your previous two rounds, introduce yourself, how was your time at ISI and IIT Kgp.

Interviewer: Gave a question on probability. Follow up question on conditional probability after that.

Me: answered

UBER

Interviewer: Puzzle where 2 cars are driven 100 kms, govt has brought in a regulation to reduce fuel consumption and has proposed two technologies X & Y. X is applied to car A and its mileage increases from 50 km/L to 75 km/L. Y is applied to car B and its mileage increases from 10 km/L to 11km/L. Which technology should be chosen and why.

Duration – ~10-15mins

Jayakrishna's Preparation Strategy: Started preparing since the start of Semester 3, Regression from ISLR, Time series from hyndman, U Dinesh Kumar for intuitive understanding. IIT Kgp DL course material for all DL related topics. Try reading from the source slides in Kgp sem itself, not recommended after you start sem3. Some source DL content: Stanford courses, UC Berkeley DL slides, Georgia Tech slides.

Blinkit

1. Robin Singh

It was an online round with different panels. Following ques were asked me in first round:

- a. Tell me about yourself
- b. Favorite Project - Regression
- c. Scenario - Blinkit has data of bills and transactions? What can we do with this data?
Business Opportunities and leveraging the benefits of business analytics
- d. Any questions you want to ask?

General Advice:

- a. Business Cases is extremely important in all companies
 - b. Practice SQL
-

2. Konda Yashwanth Goud

Round 1:

Completely Technical Round; they ask all about ML.

There was no case study.

All basic questions about ML like regression, classification, and their metrics.

If you have a good understanding of ML then it will be easy

Round 2:

It was a Technical round.

They asked a case regarding forecasting of demand.

They asked to share the screen and asked to solve SQL question in HackerRank

Blinkit

Round 3:

They again asked SQL, puzzles.

This company will grill you in ML, case studies, Puzzles, and mainly SQL.

I was not too strong in SQL. Start practising SQL from Hacker rank

Indus Insights

1. Abhishek kanwal

Test (Written Guesstimate +Math's)

Interview round 1

(Case round) Which parameters are considered for granting loans in the US market

2. Wankhade Tejas Suresh [Joined]

Resume shortlisting and a test was held. Test included a guesstimate.

(The guesstimate was to calculate average sleep hours for a society)

INTERVIEW Round 1:

1. It was a Case Study Round:
2. Case was:
You are working for a client which is a US based firm 'Quick Loans'. They provide installment home loans. You have to help in setting the price (the rate of interest)
3. Initially very little information is provided. In such cases you have to build up by asking the right question to extract that information from the interviewer and then using a structured approach arrive at an actionable solution for the client.
4. After I provided my solution ,
Q. If you have to decide between two rates of interest for this how would you approach?
5. Comparing two situations what factors do you take into account and the importance that you give them was probably evaluated here

Indus Insights

INTERVIEW Round 2:

1. It was Behavioural/CEO Round:
 2. Introduce yourself.
 3. Why Consulting?
 4. Why did you pursue your BE from University of Pune? Did you not write JEEu?
 5. When did you work very hard (instances from life)?
 6. Your Weaknesses?
 7. About Family
-

USV

Robin Singh [Joined]

The process involved two rounds. Years of experience appeared to be a criteria for shortlisting.

INTERVIEW Round 1:

- a. Tell me about yourself;
- b. Favorite Project - Regression - Why did you not consider dynamic pricing? what would have happened if considered?
- c. How does this apply in Pharma company
- d. Why PGDBA after Civil?
- e. How did you fund this program & Fee..
- f. Tell me about your family background
- g. Why PGDBA after civil again.
- h. Where did you come from?

INTERVIEW ROUND 2:

- a. Tell Me About Yourself
 - b. Why PGDBA after Civil?
 - c. Generic Que. - What are your queries regarding the company?
 - i. Robin asked : How does the pharma business run?
 - d. Tell me about your family
-

1. Abhishek kanwal

Candidates with UG (NIT and IIT) shortlisted for 2nd round

Two Interview rounds

He just wants to know if you'll stay with the company and not leave midway. Convince him that you won't reject the offer.

Sun Pharma

1. Diwakar: [Converted]

3 panellists in 1st round - One PGDBA senior batch 6, MD supply chain, HR guy.

Interview Round 1

1. Brief introduction
2. Domain-related question - what did you learn from Boston engineering
3. How power and speed are related, a question on air resistance
4. Can you fit a linear regression model if the no. of predictors is more than the data points?
5. Feature selection steps
6. Forward subset selection and backward subset selection
7. Brief about any project
8. Pick a project from your resume that you could have done better and how.
9. Questions related to work experience? What difficulties have you faced in your past work experience?

Panel 2: (IIMc alumni and one other guy)

1. Many questions focused on work experience, what type of KPIs you calculated, and how.
2. Guesstimate kinda question in the context of the work experience domain
3. Career objectives and your future plans

Tips: For Sun Pharma, Be prepared for your work experience part, and think about how you can apply your learning from PGDBA there

2. Utkarsh Sinha [Joined]

People with higher work ex were shortlisted more.

Interview Round 1 -

1. Technical + HR type of questions.
2. Told to explain the supply chain related project I mentioned in EXL work ex. More focused on the business aspects of the project.
3. Basic questions on regression / time series based on academic projects.
 - a. HR questions:
4. Tell me about a project which you felt was challenging.

SunPharma

5. Tell me something you would do differently after studying at PGDBA?

Interview Round 2:

1. Less technical than the first round. Mostly resume based and some HR questions.
Asked about some examples of different types of situations encountered in work ex.
2. Also some general questions about my magazine articles on Artificial General Intelligence.

Tips: Preparing for HR / behavioral questions helped. Also, use of pen and paper while explaining some complex projects from work ex.

3. Mayank Kale

- a. For me, the first round was a business round.
 - b. Que: Things worked or didn't work in favour:
 - i. Lacked Interest in Pharma
-

4. Avinash Kumar:

Selection criteria - work ex min 2 years, supply chain work ex better, purely analytics work ex preferred

Interview Round 1 -

4. Walk me through your CV
 5. Work ex related question like project, approach & methods used in project of work ex
 6. Ans - explained analytics project of work ex
 7. Asked tech question on cv
 8. Sql question - employee salary table given you have to calculate second highest salary where empid and salary given
 9. ROC and Aoc CURVE - Plot Roc and Aoc and asked better line between both
 10. work ex - job role, asked whether any exp in supply chain
 11. Forecasting time series -
 12. Adf test , stationary time series, MAC and MAPE difference , when to use which on,
 13. MSE, RMSE
-

SunPharma

5. Mukul Mundle [Converted]

(B.Tech + M.Tech in Chemical Engineering from IIT Kanpur, graduated in 2018. Worked for 3 years as Assistant Manager - Digital Transformation at Dr. Reddy's Laboratory. Thereafter, worked for 1 year as Business Strategy Manager - Digital Excellence at Jubilant Group)

Round 1:

They asked questions mostly on work experience. Since my work experience was similar to the role offered here, these questions were very relevant

They asked what different kinds of projects I had done while working with DRL, what kinds of challenges I had faced, and any kind of operations/manufacturing issue related to different stakeholders that I had handled.

Since this role required reporting to the CEO/MD, there was some general conversation along these lines. They asked some questions to check my knowledge of the pharmaceutical sector. I explained about different pharmaceutical unit operations during the interview. There were questions on conflict management too (since this role required interaction with senior management). [I answered that I was in similar situations and similar roles in my previous companies, hence I was aware how to deal with such scenarios]

Round 2:

This round seemed more like a formality, since they had probably already made up their mind to select me. They asked questions like "Tell us something about your family, who all are there in your family, where is your sister studying, etc."

Mukuls's Preparation Strategy: I would suggest preparing a small group of ~3-4 guys. I had made a group with 2 other students, and we used to discuss different topics and case studies. Once we came to IIM, we made this group and started preparing for this interview. IIM A, B, C casebooks are good resources to prepare for case studies. ISLR is also a good source (I had read the summary of ISLR actually!). UDK is also a really good book (better than ISLR imo). Towards DataScience and Medium articles are very helpful (preferably take tds subscription). Use ChatGPT for clarifying different algorithms, prepare HR questions' answers.

SunPharma

Prepare Graph Neural Network for Providence.

6. Saurabh Kumar

- Business round. No case studies.
 - Questions asked were all from my work ex and situation questions. My achievements in my work ex walkthrough.
 - HR questions such as why PGDBA after working in a core company for 6 years.
-

Kotak

1. Abhishek Garg [Joined]

Round 1:

Based on Resume. Started with my introduction, then asked me to explain one project and questions on that project. No case study was asked in this round, only sql and ML concepts.

Round 2:

Business round and questions related to HR and Work experience.

Axtria

1. Abhishek Kanwal [Joined]

Test + case study PPT submission for round 2

Interview round 1:

Question on Case study PPT , Simple SQL query and one segmentation project
(45 min)

Interview round 2:

1. Round with the Partner
 2. Why do you want to join the company (20 mins on this)
 3. Do you have any plans for higher study
 4. Duration of interview - 35 min
-

Accenture

1. Robin Singh

The interview process had 3 rounds including a test. Resumes were Shortlisted for Test (15-16 People). And, 8-9 people were selected after the test.

The 1 Hour (Full MCQs) test involves:

- a. SQL Coding - 5 Question
- b. Aptitude Test

Tip: Go through Coding MCQs of SQL

INTERVIEW Round 1:

- a. Tell Me About Yourself
- b. What is Your Favorite Project - Regression project was explained
- c. Time Series:
 - i. Explain ACF, PACF intuitively not technically
 - ii. Prepare answers in layman terms.
- d. What are you looking for in this internship? Expectations.
- e. How do you approach a given problem statement - Model
- f. How will you implement a model in a business problem? - Using a research paper
- g. Working of random forests - intuitively

INTERVIEW Round 2:

- a. SQL queries were asked in this round:

General Advice:

- a. Master Business Cases, SQL
-

Accenture

2. Tejas Wankhede

Interview Round 1:

1. Introduce yourself
 2. What is linear regression?
 3. What are the limitations of linear regression?
 4. Why not polynomial regression instead of non linear models?
 5. How do you make the distribution normal?
 6. Does log transformation really make the data normal? If yes how?
 7. What is adaboost? How is splitting done? Explain with an example
 8. What is gradient boosting ?
 9. How do you read CSV file? How to not make it take first row as header?
 10. Is it necessary to convert categorical columns into numerical in scikit learn?
 11. If you have two categorical columns and a numeric column, how would you take summary of it?
 12. What is normal distribution?
 13. Why do we assume distributions to be normal?
 14. What is window function?
 15. Write queries on join. Left join, right join and Inner join.
 16. What is p value?
 17. What is z score?
 18. How is p value related to z score?
 19. What is t test? What is z test? What is the difference between them? When to use which test?
 20. What is the confidence interval?
 21. -3 to +3 range for z score what is it called?"
-

3. Mukul Mundle (B.Tech + M.Tech in Chemical Engineering from IIT Kanpur, graduated in 2018. Worked for 3 years as Assistant Manager - Digital Transformation at Dr. Reddy's Laboratory. Thereafter, worked for 1 year as Business Strategy Manager - Digital Excellence at Jubilant Group)

Round 1:

Accenture

They were asking random questions (not based on our CV)

What's a bias-variance tradeoff? Then, they gave some information on customers and relations with different customers, and I approached this problem statement using graphs. He then asked me to write the python code for my solution on paper (basically we had several numbers and they were connected with each other. I had to find the maximum number connected to each number). I used 2 for loops to write the python code.

They asked me to calculate 396×47 , without using pen and paper. I approached it as $(400-3) \times (50-4)$, and gave the correct answer.

Round 2:

Started with my introduction. Then they asked me about one of my work experience projects: It was a data science project in the pharmaceutical industry, where we wanted a good yield of the products. In the manufacturing setup we'd have different unit operations' initially we'd use certain raw materials - active pharmaceutical ingredients, supporting ingredients - and finally we'd get the tablet. Yield is output/input, and we had to maximize yield. Since these raw materials are going through different unit operations, there are different parameters included in this product supply chain, parameters like material attributes, process attributes, human attributes (which person is operating which machine, temperature and pressure of the machines, etc.), and quality parameters. This was basically a regression problem. I explained the data sources, different variables involved in this problem statement and how I approached this problem statement. While I was explaining my project, they asked questions on various ML techniques: "What's random forest, what's a decision tree, what's XGBoost, etc?" I explained that I had used Shapley values and the related feature engineering part of my project.

They asked "Why Accenture?" I answered that I already had experience in the analytics field and I wanted to work in a similar environment, since I was working effectively as an internal consultant and providing analytics solutions. Accenture would give me an opportunity to interact with external clients, thereby being a natural work progression.

They enquired about my interest in adventure sports, and hobbies in general.

Accenture

4. Gaurav Uttam [Joined]

Round 1: (with Senior Principal)

1. Intro,
2. Our native location discussion as we both were from almost nearby cities (kanpur & farakkhabad) gave a +ve sign
3. Customer segmentation work ex project discussion
4. Regularisation techniques
5. Class imbalance methods & its merits/ demerits
6. Random forest

Round 2: (with MD)

1. Intro
 2. End to end an analytics project flow overview
 3. Work ex point detail discussion (non tech point)
 4. Extra-curricular point discussion
 5. Case comp. point discussion
 6. Detail discussion on my hobbies / non acad interests like cooking, book reading, my lovable sport -squash game
 7. Strength, weakness, biggest achievement so far
 8. She was interested in my explanation/interest + wanna have only elaborative discussion on each non tech points mentioned in CV specifically
At last:
 9. I asked about intern joining dates + location preferences to show my enthu in her company
-

Bloktrek

1. Nikhil Garg [Joined]

1st round was based on resume. Well prepare with your resume each point . They can ask anything They ask me technical things on project. Also my work experience and my role

Interview Round 1:

It is whole technical round for 40 minutes on deep learning since the company is based on generative AI so MY INTERVIEW IS HARD CIRE TECHNICAL.

Schneider Electric

1. Avinash Kumar [Selected]:

1 round only resume based

Interview Round 1:

1. Workex related question, job role, how analytics used in work ex, how your work can be applied in schneider, project description & highlighted term
 2. ML question on resume and cv,
 3. Regression project - regression and classification diff, scenario in which regression used
 4. NLP application
-

EXL

1. Vivekananda Chaudhary [Joined]:

There were 2 online rounds for EXL.

Interview Round 1:

1. The first round was about 15-20 minutes and it started with HR questions like introduction, about my previous work experience. Then the 2 panelists gave me Sql problem (moderate level) which could be solved using self join - I dont remember the exact question
2. Then they gave a probability question that could be solved using bayes theorem. If I remember correctly, the question was that a couple had 3 children. What is the probability of having 2 boys given that the elder was a boy or something.
3. Then they asked me some basic ML and python pandas and numpy questions. In ML, they asked me about classification errors.

That was about it in the first round.

If we have decent workex, they will go deeply in the work ex points mentioned in the CV.

Interview Round 2:

1. In the second round, the selection was based on 1 case study question, common to all the 3 remaining candidates including me. You are a Marketing Analyst in ABC bank. You have the email addresses of 1000 potential credit card customers. How to form suitable targeted marketing promotion campaign for maximum conversion?
 2. Tip - Communicate your approach at regular intervals to the panelist (there was a single panelist in the second round as opposed to 2 panelists in the first round). Ask questions in between to the panelists to ensure the correct direction of thr case study.
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2. Wankhade Tejas Suresh

There were 2 online rounds for EXL.

Interview Round 1:

A guesstimate was asked:

1. Estimate total no of cars sold in Mumbai in 2022.
2. Some question on applications and use cases of SQL concepts (Window functions etc)
3. Your previous coding experience and which languages and libraries you've worked on?
4. What are LLMs?

EXL

Interview Round 2:

A business case study was asked in this round

Problem statement:

1. You are working for a client which is a bank. You have a list of 10000 people to be targeted for Credit Card sales. You have to come up with a email communication strategy for maximum conversion (You do not have their contact details only email ids are available with us)
 2. What is your current CGPA?
-

Auxo AI

1st round was theory based.

2nd is to see if you code as pgdba batch 4/5 alums took the first 2 round.

The third round was taken by MD.

Interview Round 1:

The interview was heavily focused around Deep Learning projects. I had BERT project on my resume.

1. I was asked about tokenization in BERT, explain how a word piece tokenizer works (on paper). What are the other kinds of tokenizer that can be used?

2)How can you get output from all 12 hidden states(encoders) ?

3)Questions around architecture of BERT. Follow-up questions on that. How do you train a BERT for a

4)Question Answering task? Explain the steps involved

5)How is Sentiment analysis done using bert?Explain the steps involved.(I told the max pooling and CLS based approach)

6)Why do we use skip connections(Often due to vanishing gradient the entire layers below a certain layer will not learn anything. In order to prevent this skip connections helps provide information flow through alternate channel, to bottom layers hence improving learning for the network)

7)If it is not stationary, what problems can occur in AR/ARIMA/SARIMA modeling?(If it is not stationary, then the variance is not constant, which means prediction intervals, confidence intervals, and the entire analysis will be inaccurate. Moreover, if the expected value is not constant, some predictions will also be incorrect.)

8)Word2Vec and it's type

9)What is Glove and when do we use it.

35 mins Rd

Interview Round 2:

WEX based question.Steps of deployment.How did you deploy your project on cloud using AWS.(10-15 min discussion).How will you use LSTM/Bi-directional LSTM for time series(Tell in Tensorflow/Pytorch).I told them about time distributed layer of Tensorflow .Parameters of XGB/Random Forest .30 mins rd

Interview Round 3:

In Third round Guesstimate was given(amounts of bread sold in India).Along with that Wex projects were asked and what business problems do they solve.What else could you have done.Discussion on Medical Charges Prediction(Regression).The interviewer was more interested in the business statement of project.25 mins rd

Resources:

Wes McKinney - Python for Data Analysis_ Data Wrangling with pandas, NumPy, and Jupyter- O'Reilly Media (2022)

Aurélien Géron - Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow - Concepts, Tools, and Techniques to Build Intelligent Systems, 2nd Edition (2019)

CS-182N(Berkley Videos)

mitish khapra Video for Word2vec/Glove Etc